Electrostatics and Electric flux

Sat May 09 2026

Welcome

Get ready to give your best





Am I am audible
and
visible to you ?



If there is any audio or video issue thenrefresh.
Do check internet Speed



If nothing works out simply logout of session and rejoin





Electrostatics

Grade 12 • Physics • Session 3



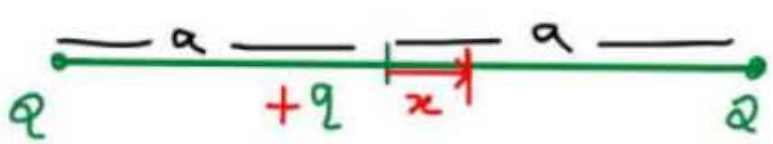


Vedantu
Learn LIVE Online

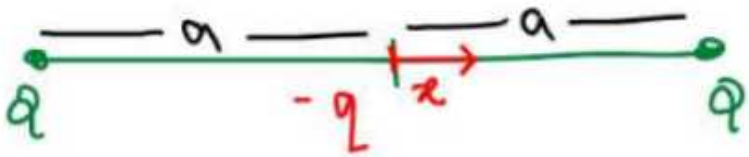
Page 14

$F \propto -x$

$x << a$



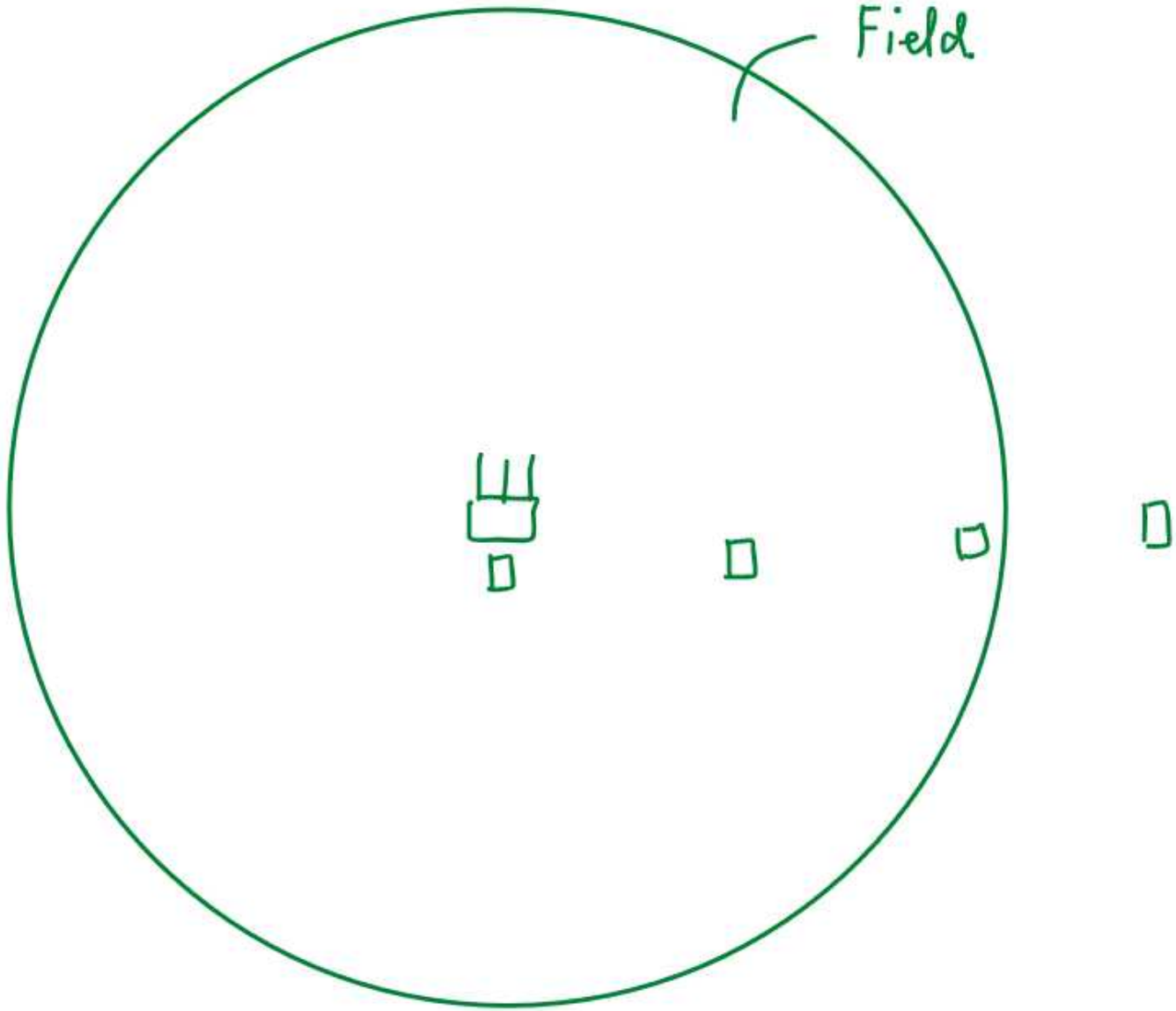
$F \propto x$

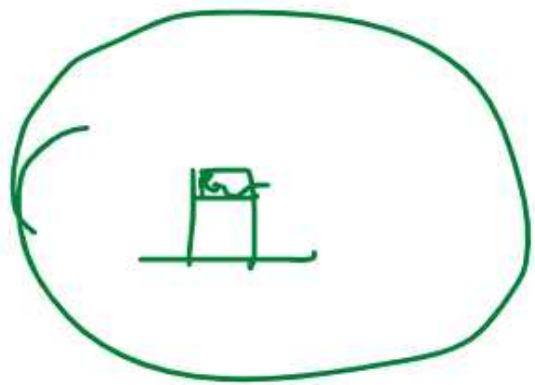


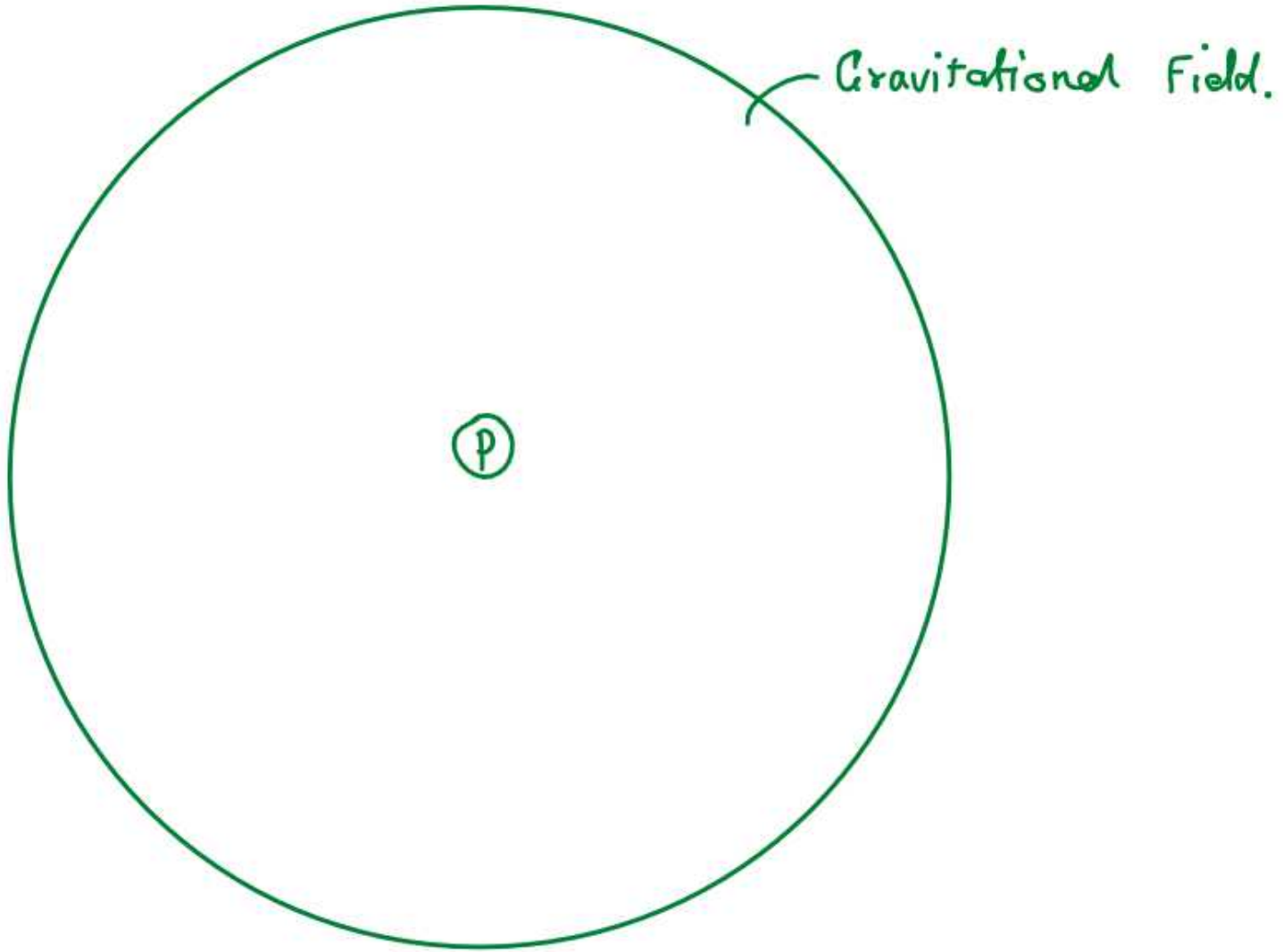
Recap

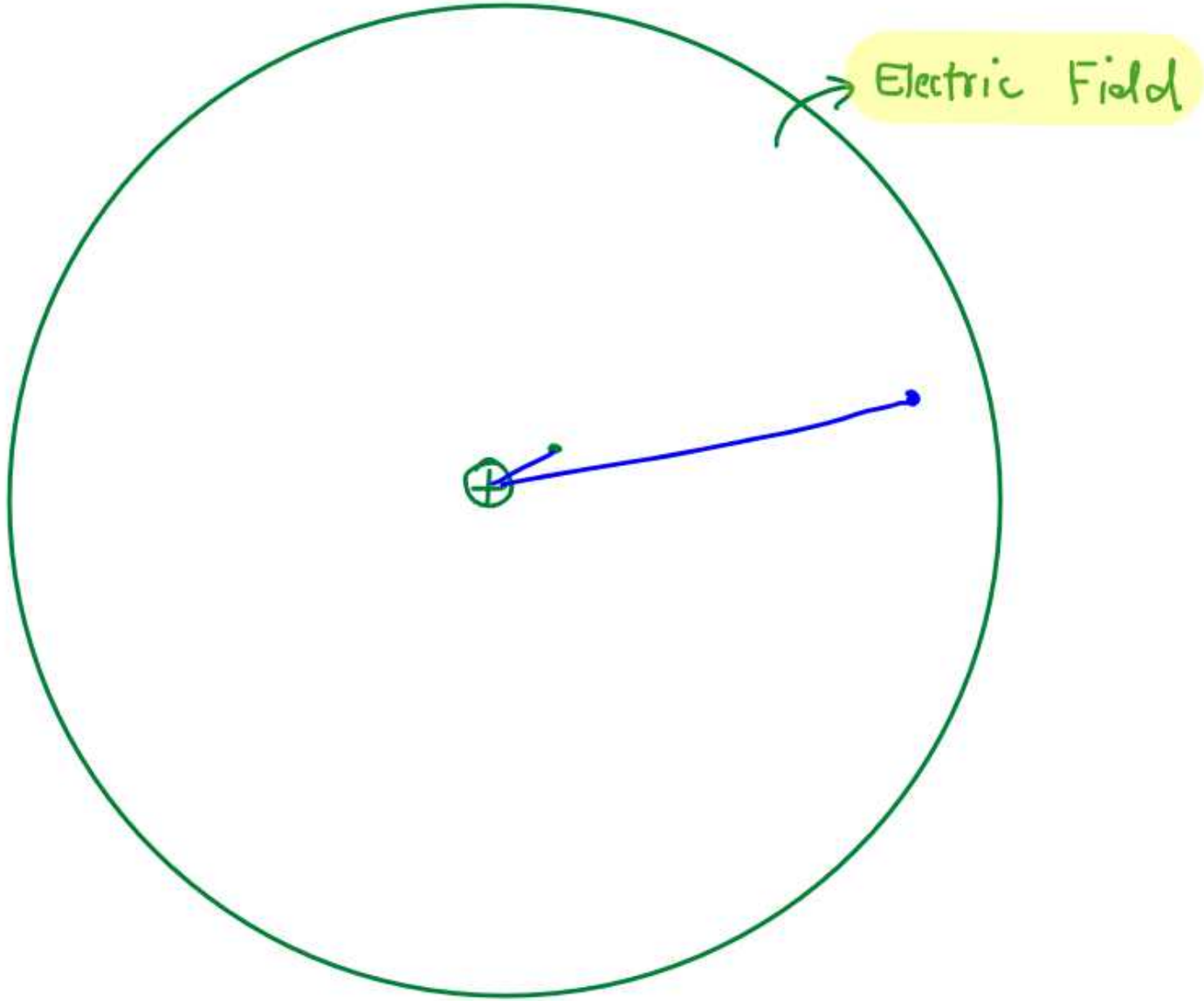
- Introduction to Electrostatics
- Electric Charge
- Properties of Electric Charge
  - Additivity
  - Conservation
  - Quantisation
- Conductors and Insulators
- Methods of charging a body
  - Friction (rubbing)
  - Conduction
  - Induction
- Coulomb's Law

Vedantu
Learn LIVE Online

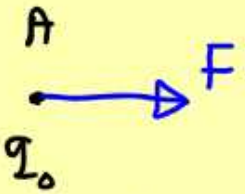

Field




Gravitational Field.
P


Electric Field

Electric field Intensity ($\bar{E}$)

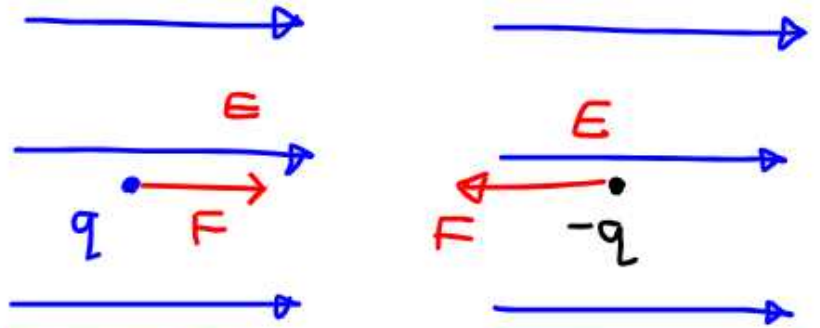


(Test charge)

$$\bar{E} = \frac{\bar{F}}{q_0}$$

Vector quantity

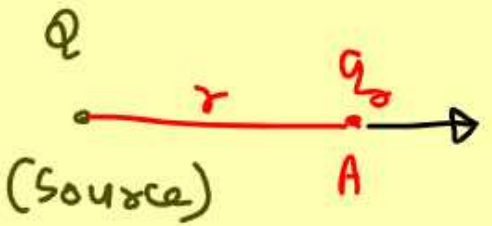
Unit: N/C, V/m



$\bar{F} = q\,\bar{E}$

$\bar{F} = -q\bar{E}$

E due to a point charge.



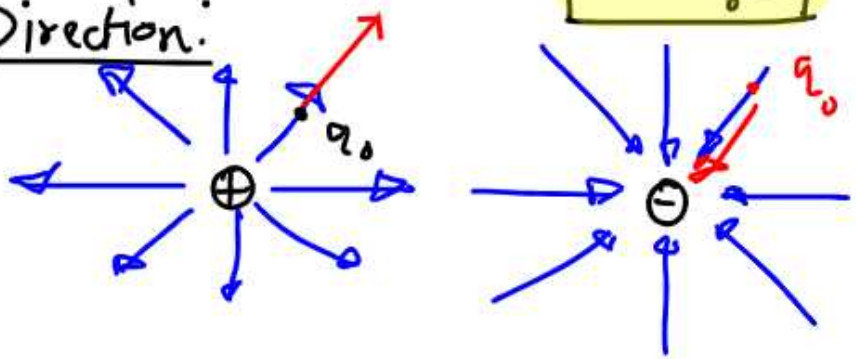
$$F = \frac{kQq_0}{r^2}$$

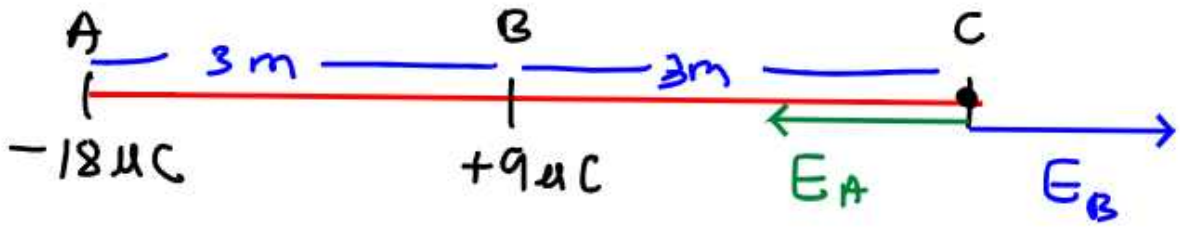
$$E = \frac{F}{q_0} = \frac{\frac{kQq_0}{r^2}}{q_0}$$

$$E = \frac{kQ}{r^2}$$

$k = \frac{1}{4\pi\varepsilon_0}$

Direction:





E_{Net} @ C. = ?

−ve ⟷ +ve

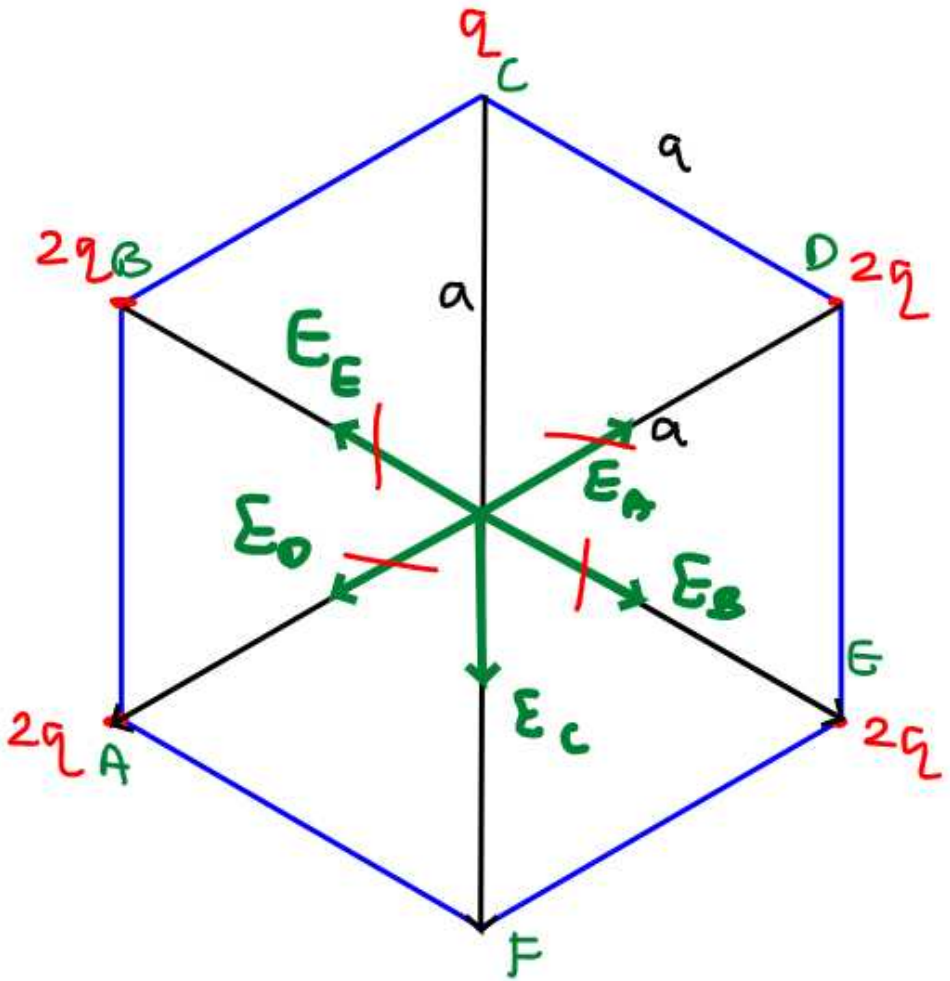
$$\bar{E}_{Net} = +E_B - E_A$$

$$= \frac{k(9)\times10^{-6}}{3^2} - \frac{k(18)\times10^{-6}}{6\times6}$$

$$= k\times10^{-6}\left(1-\frac{1}{2}\right)$$

$$= 9\times10^{9}\times10^{-6}\times\frac{1}{2}$$

$$= 4.5\times10^{3}\ N/C\ (\rightarrow)$$



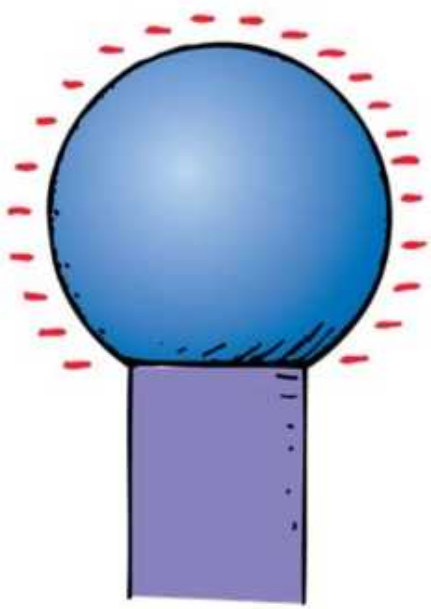
$$E_E = E_D = E_B = E_A$$

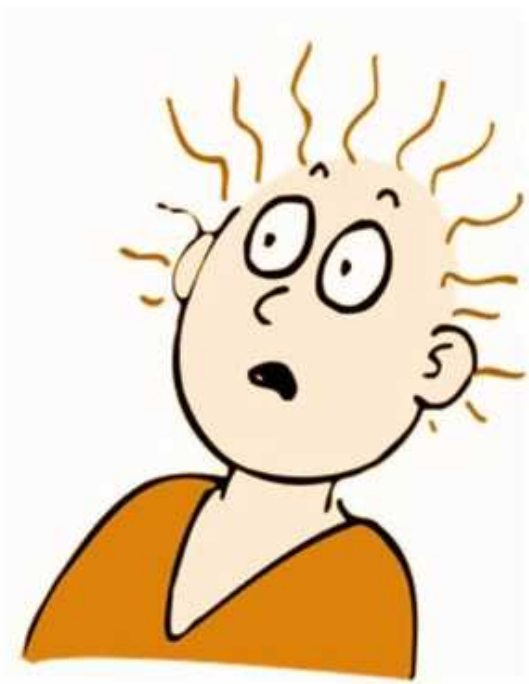
$$E_{Net} = E_C = \frac{kQ}{r^2} = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{a^2}$$

Electric field

If you walk by the charged dome of an electrostatic machine (for example a Van de Graaff generator), you can sense the charge. Hair on your body stands out—just a tiny bit if you're more than a meter away, and more if you're closer.

The space is said to contain an electric force field (or just, electric field).





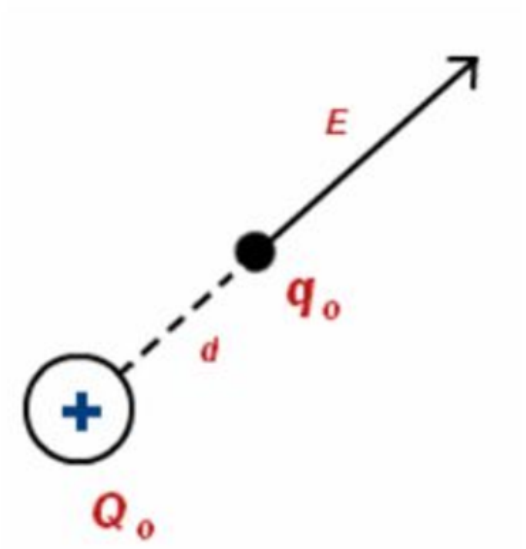
Vedantu
Learn LIVE Online

Electric field

Intensity of Electric Field

The Intensity of Electric Field E at a point in space is defined as the force Fe acting on a very small positive test charge q0 placed at that point divided by the test charge

$$\vec{E} = \lim_{q_0 \to 0} \frac{\vec{F}}{q_0}$$



Electric Field is a vector quantity

Units and Dimensions of electric field

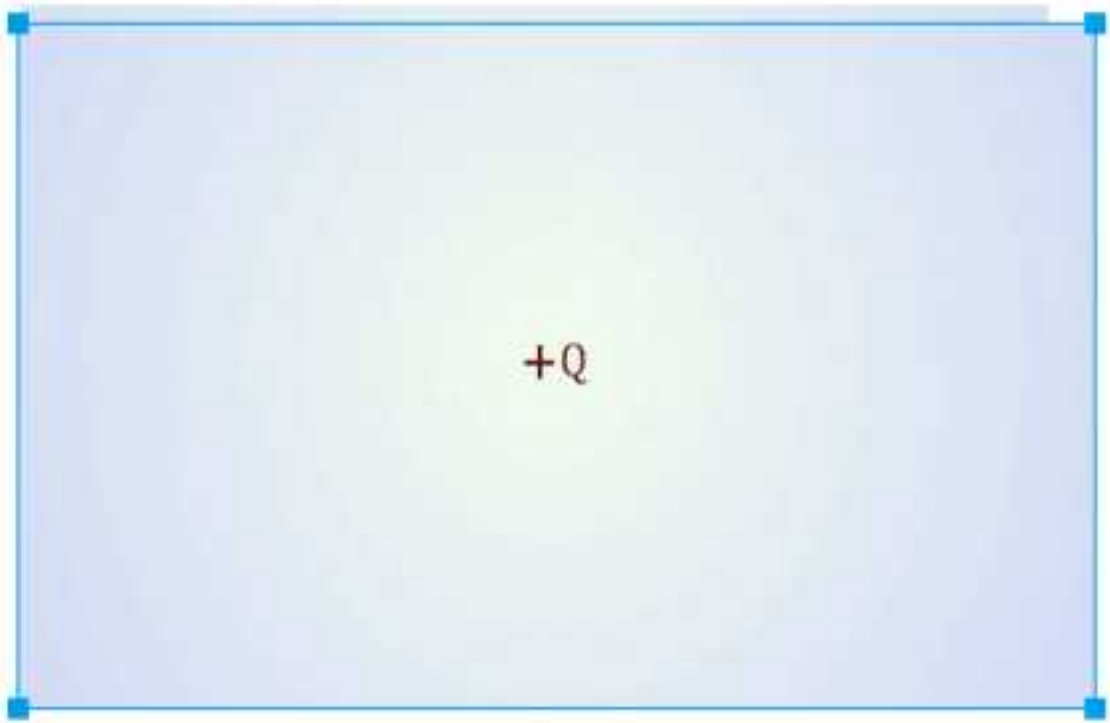
SI Unit : NC^{-1}

$[E] = [MLT^{-3} A^{-1}]$

Vedantu
Learn LIVE Online

Electric field due to a point charge

Let a charge + Q is placed in space and we have to find the electric field at a distance r from it.



Mathematically,

$$\vec{E} = \frac{\vec{F}}{q_0}$$

And we know, $\vec{F} = \frac{1}{4\pi\varepsilon_0}\frac{Qq_0}{r^2}\hat{r}$

So, $\vec{E} = \frac{\vec{F}}{q_0} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\hat{r}$

$$\Rightarrow \vec{E} = \frac{kQ}{r^2}\hat{r}$$

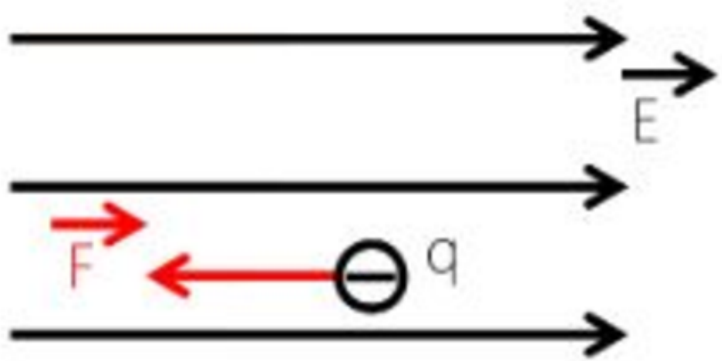
Vedantu
Learn LIVE Online

Effect of Electric Field on a Charged Particle

(1). If q is positive then $\vec{F}$ will be parallel to



(2). If q is negative then $\vec{F}$ will be antiparallel to



Vedantu
Learn LIVE Online

Effect of Electric Field on a Charged Particle

Chat Off



Interactivity Pod

A charged particle of mass 'm' and negative charge '-q' was found floating in a room. What is the direction of electric field in the room?

A downwards

B Upwards

Vedantu
Learn LIVE Online

Effect of Electric Field on a Charged Particle

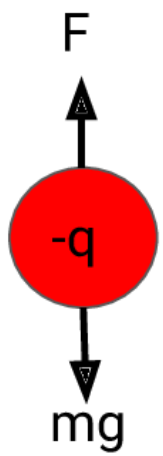
Chat On

Interactivity Pod

A charged particle of mass 'm' and negative charge '-q' was found floating in a room. What is the direction of electric field in the room?

(A) downwards

(B) Upwards



If q is negative then F will be antiparallel to E, so we can say that direction of electric field is downwards.

Vedantu
Learn LIVE Online

Let's Solve.



Vedantu
Learn LIVE Online

Lets Solve

Q. The magnitude of electrostatic force experienced by a charge of $\mathbf{1 \times 10^{-6} C}$ is $\mathbf{1.5 \times 10^{-3} N}$. Find the magnitude of the electric field at the position of the charge.

Vedantu
Learn LIVE Online

Lets Solve

Q. The magnitude of electrostatic force experienced by a charge of $1 \times 10^{-6} C$ is $1.5 \times 10^{-3} N$. Find the magnitude of the electric field at the position of the charge.

Solution:

$$\vec{F} = q\vec{E}$$

$$\Rightarrow |\vec{F}| = |q\vec{E}|$$

$$\Rightarrow F = qE$$

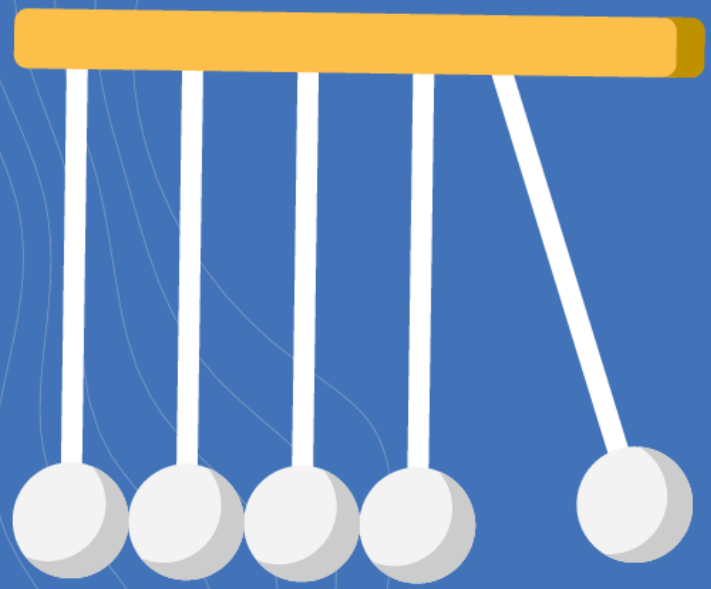
$$\Rightarrow 1.5 \times 10^{-3} = (1 \times 10^{-6})E$$

$$\Rightarrow E = \frac{F}{q} = \frac{1.5 \times 10^{-3}}{1 \times 10^{-6}} N/C$$

$$\Rightarrow E = 1.5 \times 10^{3} N/C$$

Vedantu
Learn LIVE Online

Quiz.



Vedantu
Learn LIVE Online

Quiz

Chat Off

Q. At a particular point, electric field depends upon

A. Source charge Q only

B. Test charge q_0 only

C. Both Q and q_0

D. Neither Q nor q_0

Vedantu
Learn LIVE Online

Quiz

Chat On

Q. At a particular point, electric field depends upon

A. Source charge Q only

B. Test charge q_0 only

C. Both Q and q_0

D. Neither Q nor q_0

Vedantu
Learn LIVE Online

Quiz

Chat Off

Q. What is the magnitude of a point charge which produces an electric field of $2NC^{-1}$ at a distance of 60 cm ?

$\left(1/4\pi\varepsilon_0 = 9 \times 10^9\ Nm^2C^{-2}\right)$

A $8 \times 10^{-11}C$

B $2 \times 10^{-12}C$

C $3 \times 10^{-11}C$

D $6 \times 10^{-10}C$

Vedantu
Learn LIVE Online

Quiz

Chat On

Q. What is the magnitude of a point charge which produces an electric field of $2NC^{-1}$ at a distance of 60 cm ?

$\left(1/4\pi\varepsilon_0 = 9 \times 10^9\ Nm^2C^{-2}\right)$

Solution:

Here,

$\vec{E} = 2\ NC^{-1}$

$r = 60\ cm = 60 \times 10^{-2} m$

$Q = ?$

$$E = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}$$

$$\Rightarrow 2 = 9 \times 10^9 \times \frac{Q}{(0.6)^2}$$

$$\Rightarrow Q = \frac{2 \times 0.36}{9} \times 10^{-9} \quad \Rightarrow Q = 8 \times 10^{-11} C$$

Vedantu
Learn LIVE Online

Quiz

Chat Off

$g = 10 m/s^2$

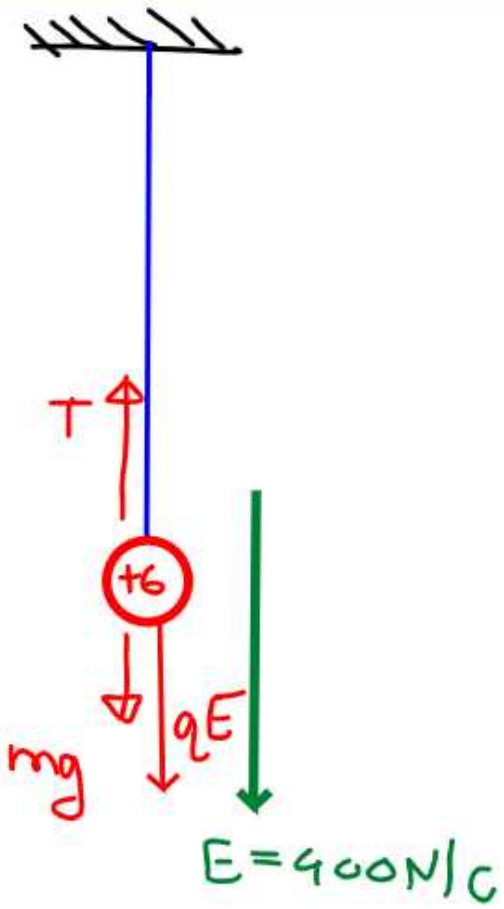
Q. A small sphere of mass 1 g carries a charge of +6 μC. The sphere is suspended by a string in an electric field of 400 NC^{-1} acting downwards. Calculate tension in the string.

A $1.24 \times 10^{-2}\ N$

B $2.11 \times 10^{-2}\ N$

C $5.11 \times 10^{-2}\ N$

D $1.24 \times 10^{-3}\ N$



$\Sigma F_y = 0$

$T = mg + qE$

$T = 1 \times 10^{-3} \times 10 + 6 \times 10^{-6} \times 400$

$T = 10^{-2} + 0.24 \times 10^{-2}$

$T = (1 + 0.24) \times 10^{-2}$

Vedantu
Learn LIVE Online

Quiz

Chat On

Q. A small sphere of mass 1 g carries a charge of +6 μC. The sphere is suspended by a string in an electric field of 400 NC^{-1} acting downwards. Calculate tension in the string.

A $1.24 \times 10^{-2}\ N$

B $2.11 \times 10^{-2}\ N$

C $5.11 \times 10^{-2}\ N$

D $1.24 \times 10^{-3}\ N$

Vedantu
Learn LIVE Online

Quiz

Chat On

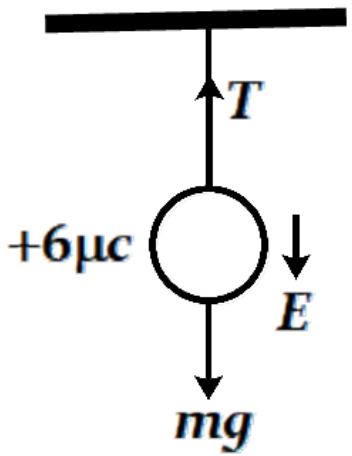
Q. A small sphere of mass 1 g carries a charge of $+6\,\mu C$. The sphere is suspended by a string in an electric field of 400 NC^{-1} acting downwards. Calculate tension in the string.

Solution:

Here, $m = 1\ g = 10^{-3}\ kg, q = +6\mu C = +6 \times 10^{-6} C$

$E = 400\ NC^{-1}$ downwards.

From figure,



$$T = F_e + mg = qE + mg$$

$$= 6 \times 10^{-6} \times 400 + 10^{-3} \times 10$$

$$= 124 \times 10^{-4} N = 1.24 \times 10^{-2} N$$

Vedantu
Learn LIVE Online

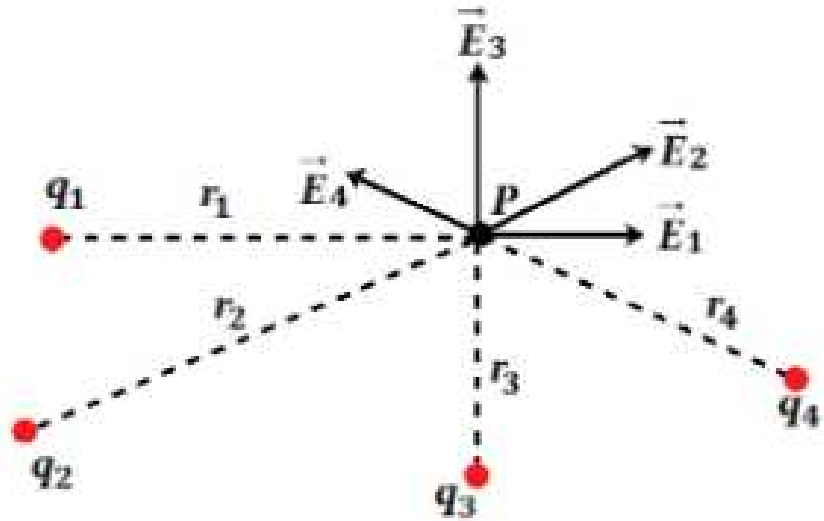
Electric field due to a system of charges

Electric field at a point due to a system of charges is the vector sum of the electric fields at the point due to all the individual charges.

$$\vec{E}(P) = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \vec{E}_4$$

$$\Rightarrow \vec{E}(P) = \frac{kq_1}{r_{1P}^2}\hat{r}_1 + \frac{kq_2}{r_{2P}^2}\hat{r}_2 + \frac{kq_3}{r_{3P}^2}\hat{r}_3 + \frac{kq_4}{r_{4P}^2}\hat{r}_4$$

$$\Rightarrow \vec{E}(P) = \frac{1}{4\pi\varepsilon_0}\left(\frac{q_1}{r_{1P}^2}\hat{r}_1 + \frac{q_2}{r_{2P}^2}\hat{r}_2 + \frac{q_3}{r_{3P}^2}\hat{r}_3 + \frac{q_4}{r_{4P}^2}\hat{r}_4\right)$$



Electric field due to a system of charges

If there are n charges, then electric field at a point due to these charges is given by:

$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \cdots + \vec{E}_n$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{q_1}{r_{1P}^2}\hat{r}_{1P} + \frac{1}{4\pi\varepsilon_0}\frac{q_2}{r_{2P}^2}\hat{r}_{2P} + \cdots\cdots + \frac{1}{4\pi\varepsilon_0}\frac{q_n}{r_{nP}^2}\hat{r}_{nP}$$

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{n}\frac{q_i}{r_{iP}^2}\hat{r}_{iP}$$

E is a vector quantity that varies from one point to another point in space and is determined from the positions of the source charges.

Vedantu
Learn LIVE Online

Let's Solve.



Vedantu

Lets Solve

Q. Two charges of 1.0 μC and 2.0 μC are placed 10 cm apart. At what distance (in cm) from the 1 μC charge (on the line joining the two charges) is the electric field zero?

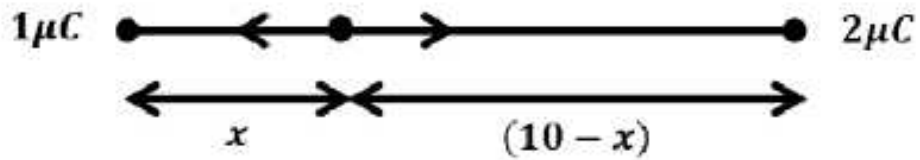
Vedantu
Learn LIVE Online

Lets Solve

Q. Two charges of 1.0 μC and 2.0 μC are placed 10 cm apart. At what distance (in cm) from the 1 μC charge (on the line joining the two charges) is the electric field zero?

Solution:

Since the electric field is zero at the given point, the magnitude of electric field must be equal due to the individual charges..



$$\Rightarrow \frac{k(1\times10^{-6})}{(x\times10^{-2})^2}=\frac{k(2\times10^{-6})}{((10-x)\times10^{-2})^2}$$

$$\Rightarrow (10-x)^2=2(x)^2$$

$$\Rightarrow 10-x=\pm\sqrt{2}x$$

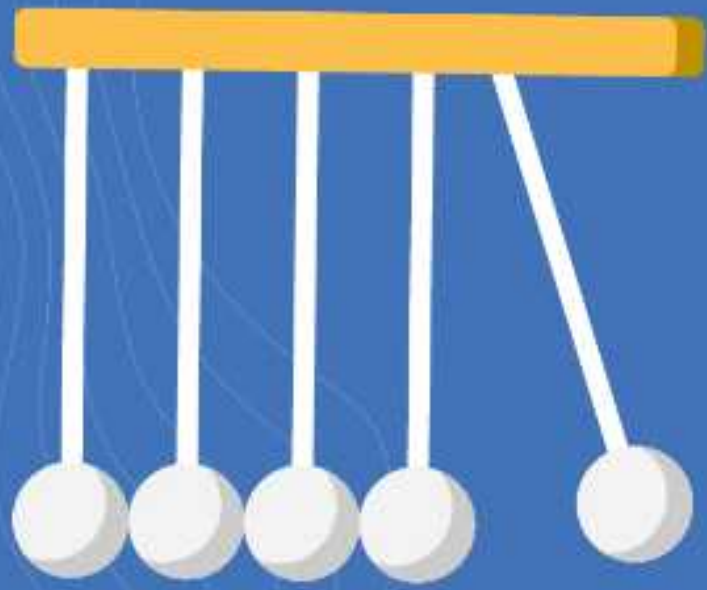
$$\Rightarrow 10-x=\sqrt{2}x \text{ or } x=10-x=-\sqrt{2}x$$

$$\Rightarrow x=\frac{10}{1+\sqrt{2}}cm \text{ (or) } x=\frac{10}{1-\sqrt{2}}cm \text{ (not possible)}$$

$$\Rightarrow x=4.14\ cm \text{ (In compliance with the rules regarding significant digits)}$$

Vedantu
Learn LIVE Online

Quiz.



Quiz

Chat Off

Q. If there are two equal and opposite charges are there at a equal distance and collinear from a point each having electric field E then the net electric field at that point will be

A. 2E

B. E

C. 0

D. -2E

Vedantu

Quiz

Chat On

Q. If there are two equal and opposite charges are there at a equal distance and collinear from a point each having electric field E then the net electric field at that point will be

A. 2E

B. E

C. 0

D. -2E

Vedantu

Quiz

Chat On

Q. If there are two equal and opposite charges are there at a equal distance and collinear from a point each having electric field E then the net electric field at that point will be

Solution:

Due to the principle of superposition and electric field being a vector quantity , the net electric field will be zero at that point

Vedantu

Quiz

Chat Off

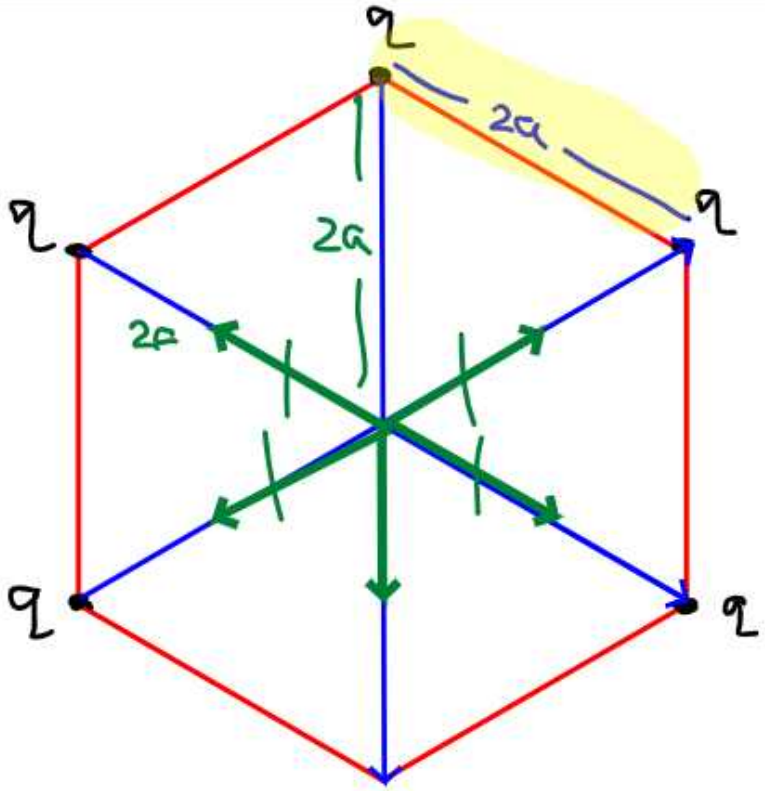
Q. Five point charges (+q each) are placed at the five vertices of a regular hexagon of side 2a. What is the magnitude of the net electric field at the centre of the hexagon?

A. $\frac{1}{4\pi\varepsilon_0}\frac{q}{a^2}$

B. $\frac{q}{16\pi\varepsilon_0 a^2}$

C. $\frac{\sqrt{2}q}{4\pi\varepsilon_0 a^2}$

D. $\frac{5q}{16\pi\varepsilon_0 a^2}$



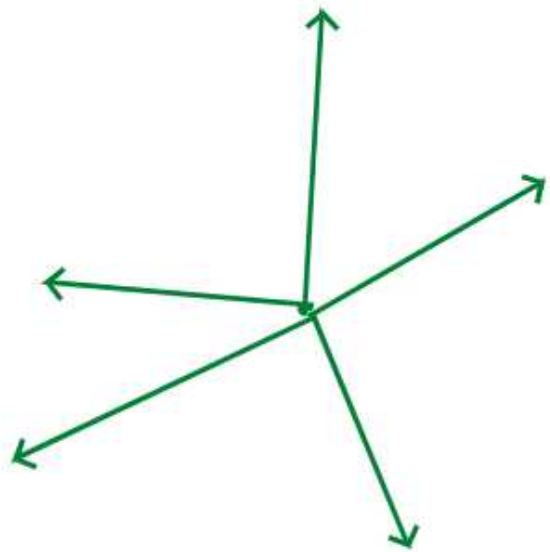
$E = \frac{kq}{r^2}$

$E_{Net} = \frac{kq}{r^2}$

$= \frac{1}{4\pi\varepsilon_0}\frac{q}{(2a)^2}$

$= \frac{q}{16\pi\varepsilon_0 a^2}$

Vedantu
Learn LIVE Online



Quiz

Chat On

Q. Five point charges (+q each) are placed at the five vertices of a regular hexagon of side 2a. What is the magnitude of the net electric field at the centre of the hexagon?

A $\frac{1}{4\pi\varepsilon_0}\frac{q}{a^2}$

B $\frac{q}{16\pi\varepsilon_0 a^2}$

C $\frac{\sqrt{2}q}{4\pi\varepsilon_0 a^2}$

D $\frac{5q}{16\pi\varepsilon_0 a^2}$

Vedantu
Learn LIVE Online

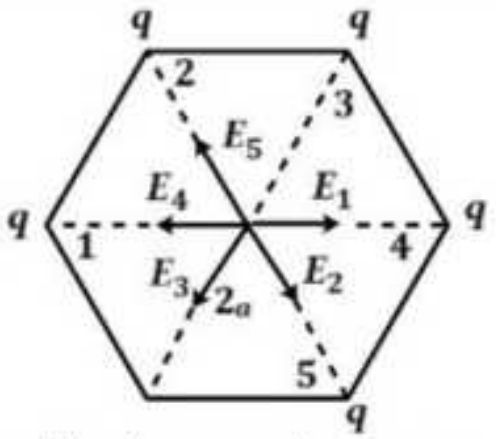
Quiz

Chat On

Q. Five point charges (+q each) are placed at the five vertices of a regular hexagon of side 2a. What is the magnitude of the net electric field at the centre of the hexagon?

Solution:

Five point charges (+q each) are placed at the given vertices of a regular hexagon of side 2a. Let the electric fields be as shown in the figure.



E_1 and E_4 are cancelled (Equal in magnitude and opposite in direction)

Similarly, E_2 and E_5 are cancelled.

Vedantu
Learn LIVE Online

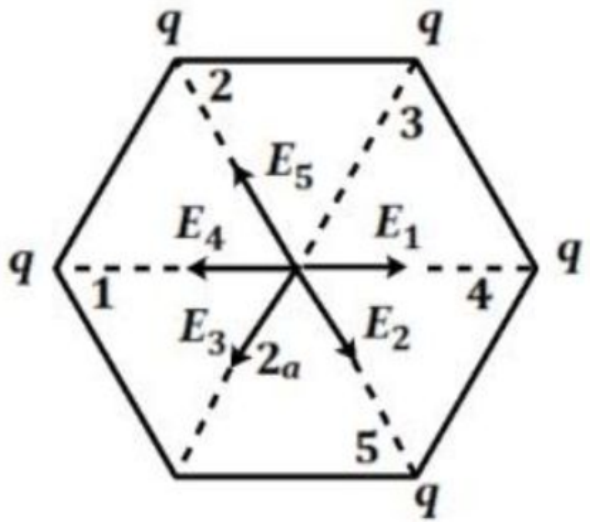
Chat Off

Q. Five point charges (+q each) are placed at the five vertices of a regular hexagon of side 2a. What is the magnitude of the net electric field at the centre of the hexagon?

Solution:

$$\Rightarrow E_{net}(\text{magnitude}) = E_3 = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q}{(2a)^2}$$

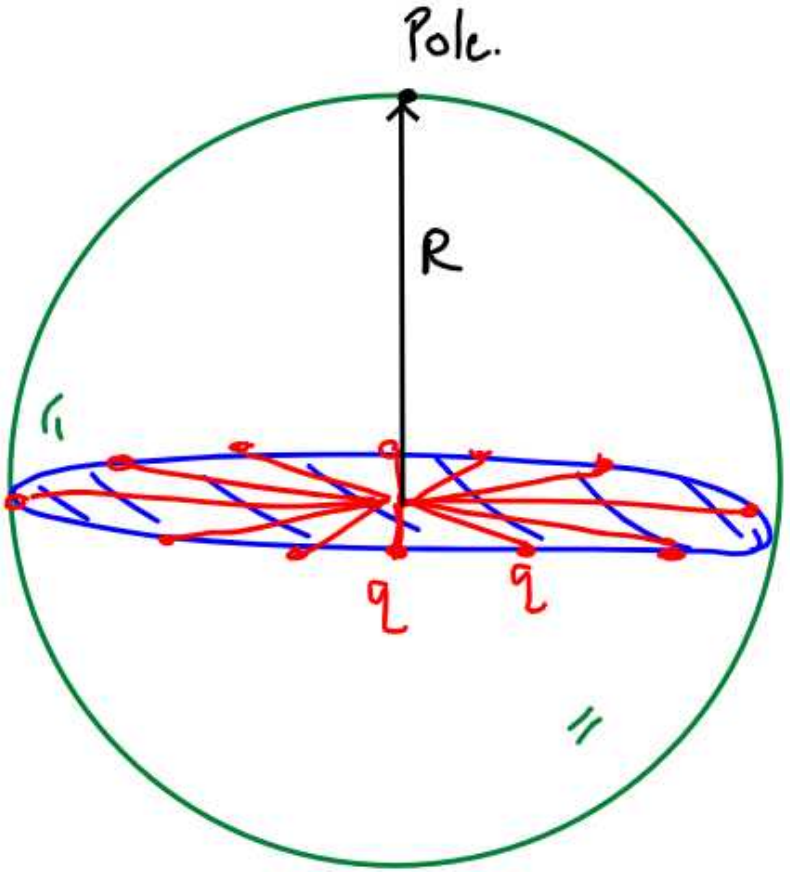
$$= \frac{q}{16\pi\varepsilon_0 a^2}$$

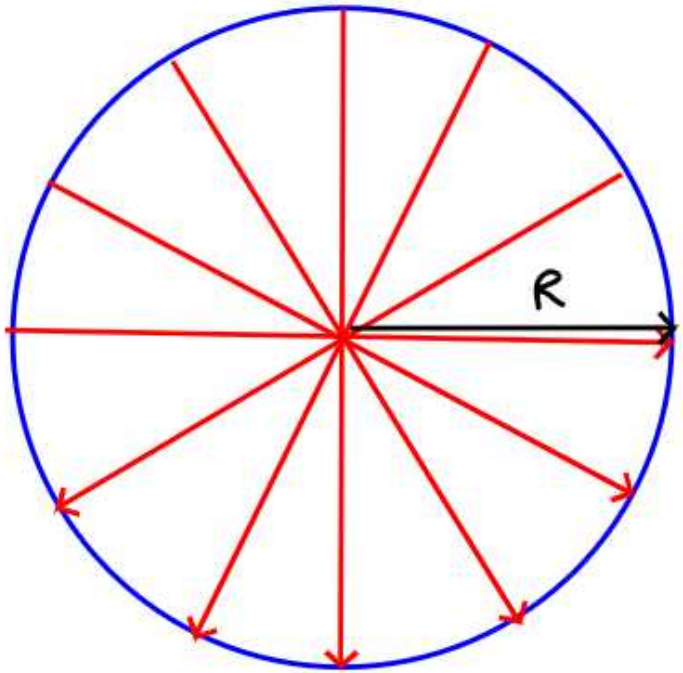


Vedantu
Learn LIVE Online

$E_{pole} = ?$

HW



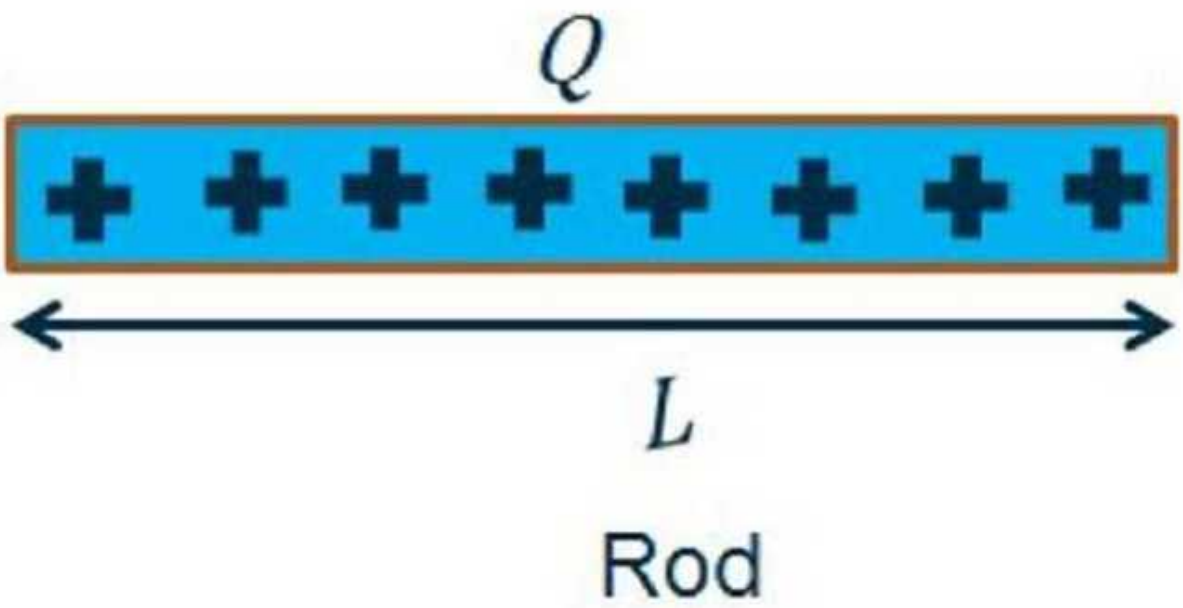


Linear Charge Density

- **Linear** charge density: when a charge is distributed along a line.

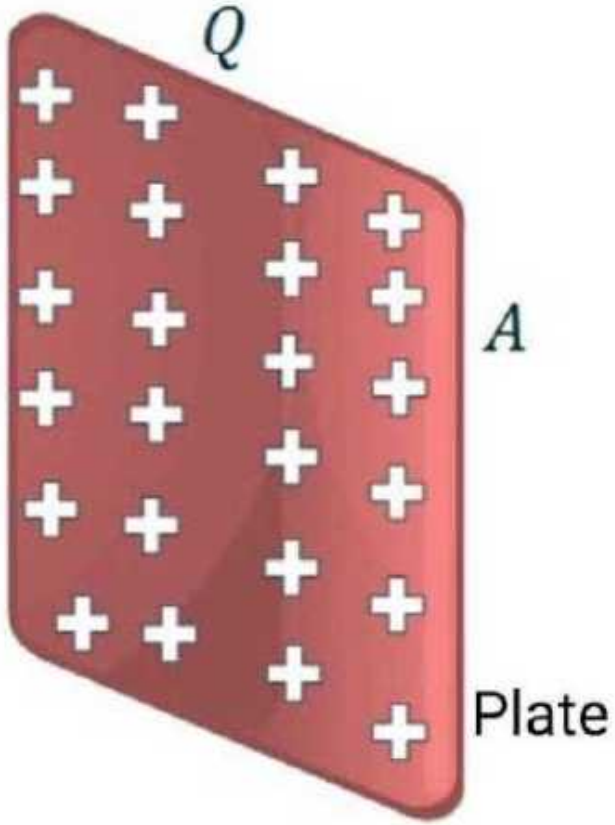
$$\lambda = \frac{Q}{L}$$

SI unit: C/m



Charge Densities

- Surface charge density: when a charge is distributed evenly over a surface area.

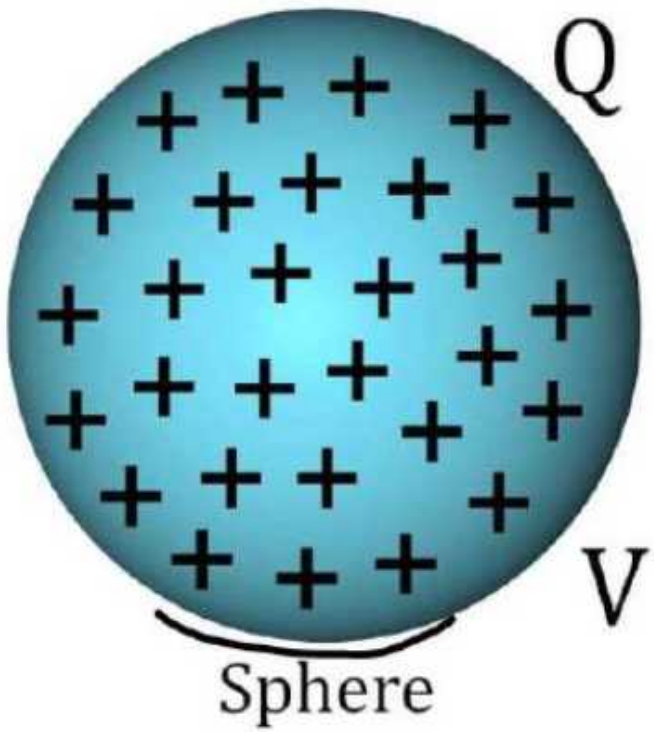


$$\sigma = \frac{Q}{A}$$

SI unit: C/m^2

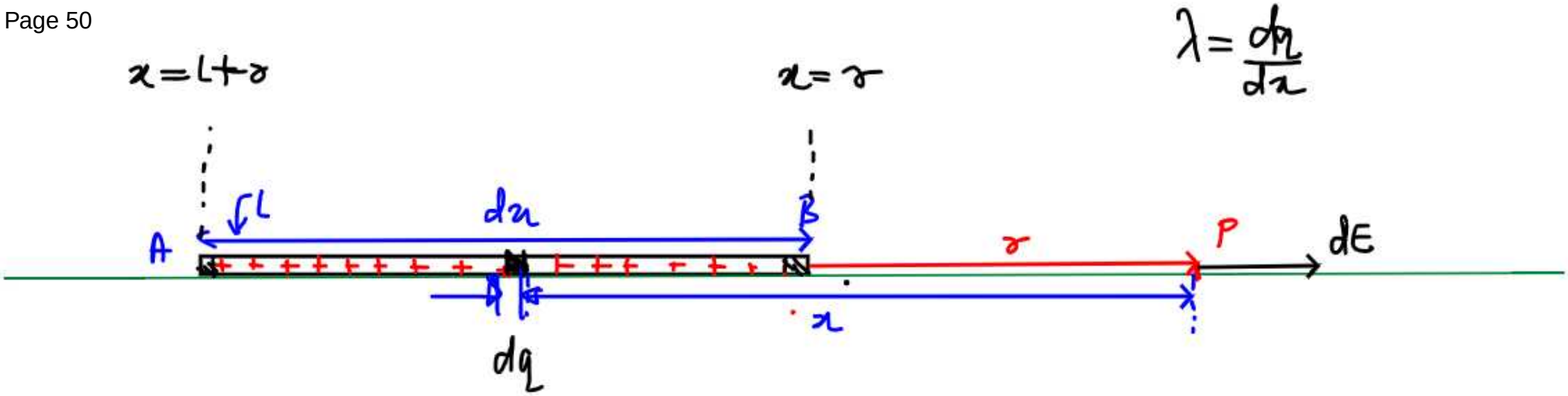
Charge Densities

- Volume charge density: when a charge is distributed evenly throughout a volume.



$$\rho = \frac{Q}{V}$$

SI unit: C/m^3



$$dE = \frac{k\,dq}{x^2}$$

$$[\, dq = \lambda\, dx]$$

$$dE = \frac{k\lambda\, dx}{x^2}$$

$$\int dE = k\lambda \int_{r}^{r+L} x^{-2}\, dx$$

$$E = k\lambda \left[\frac{x^{-1}}{-1}\right]_{r}^{r+L}$$

$$E = -k\lambda \left[\frac{1}{x}\right]_{r}^{r+L}$$

$$E = -k\lambda \left[\frac{1}{r+L} - \frac{1}{r}\right]$$

$$E = k\lambda \left[\frac{1}{r} - \frac{1}{r+L}\right]$$

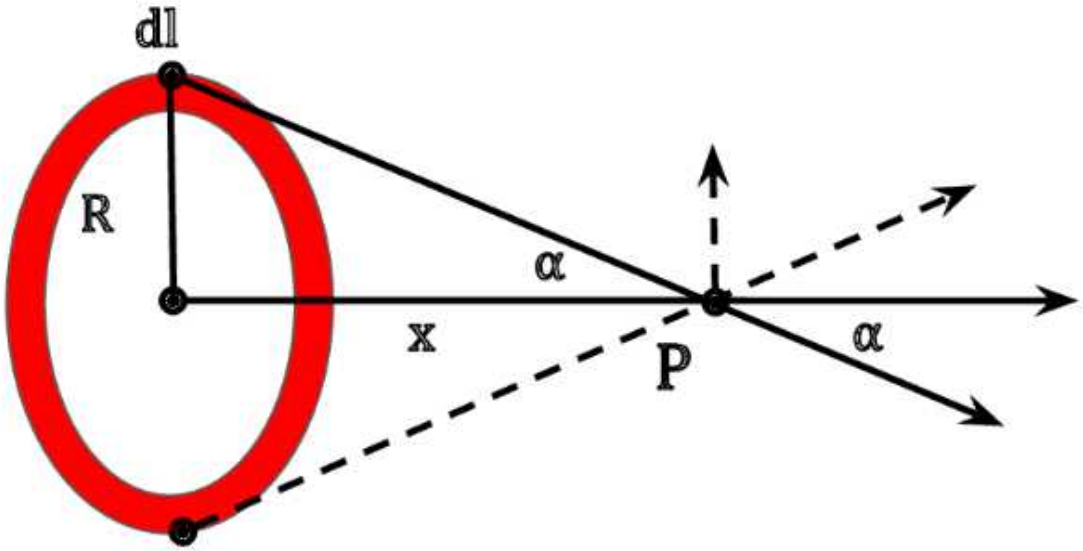
$$E = k\lambda \left[\frac{r+L-r}{r(r+L)}\right]$$

$$E = \frac{k(\lambda L)}{r(r+L)}$$

$$\boxed{E = \frac{kQ}{r(r+L)}}$$

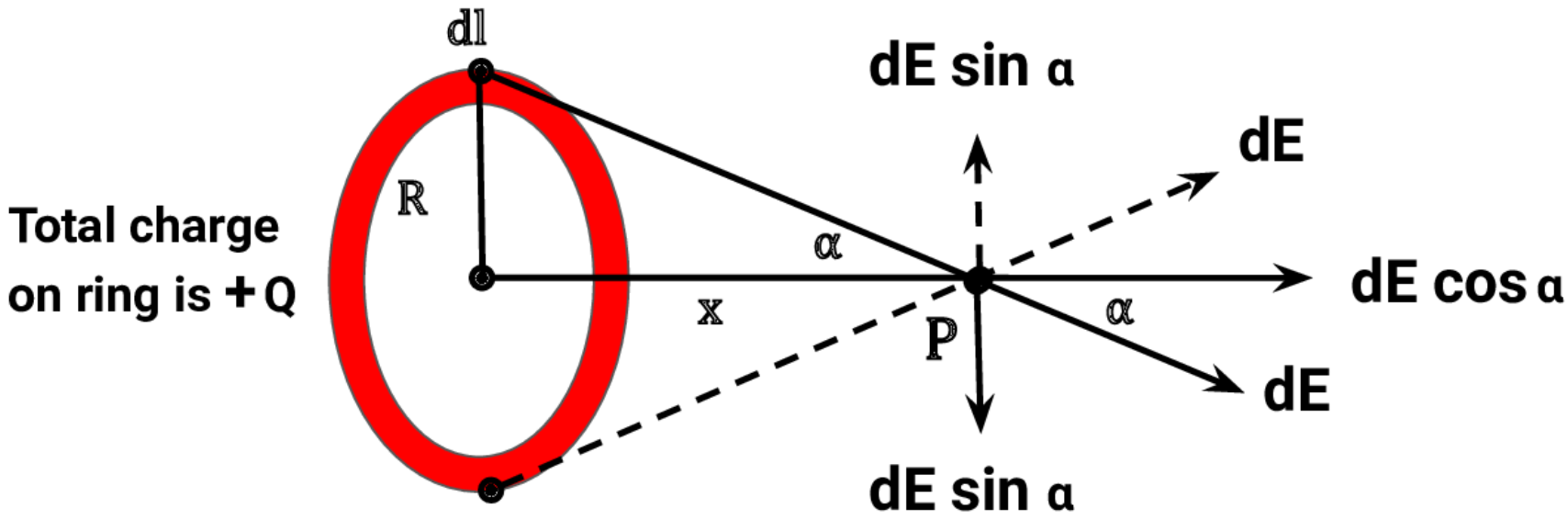
Electric field due to a Uniformly Charged Ring

Total charge on ring is +Q



Vedantu
Learn LIVE Online

Electric field due to a Uniformly Charged Ring



Point P is at equal distance from every point on the ring.

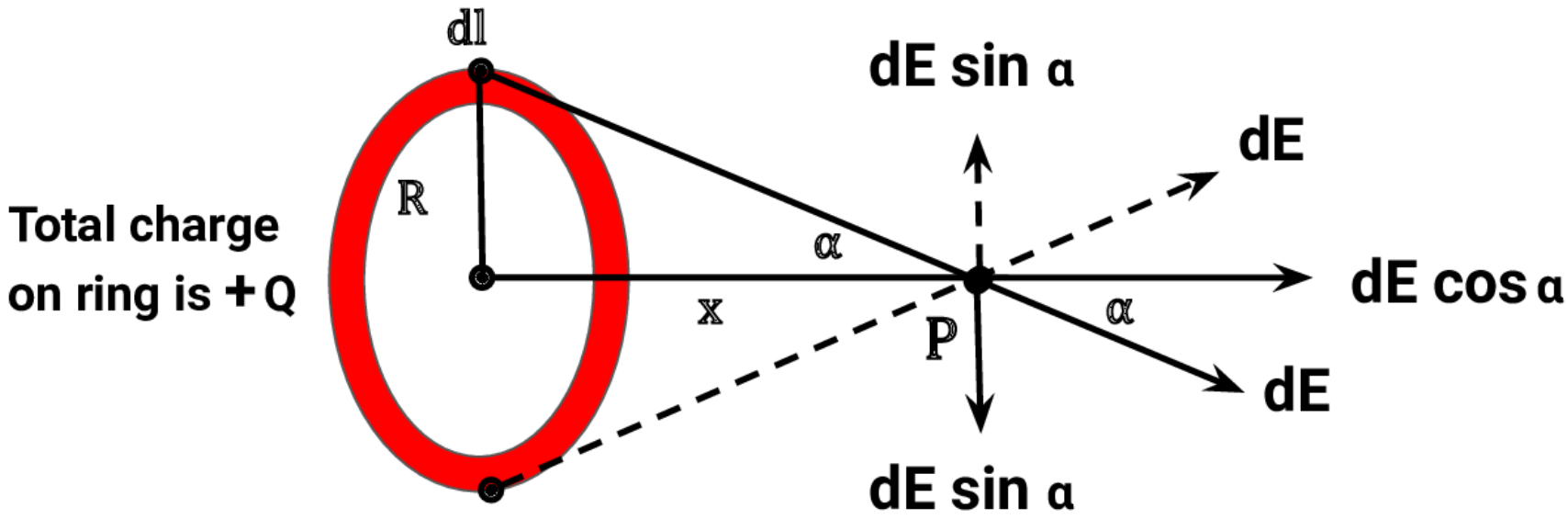
Linear charge density for a ring of radius R, $\lambda = \frac{Q}{2\pi R}$

Electric field due to a small element of length dl, $dE = \frac{kdq}{R^2+x^2}$

Where dq is the charge on small element dl. $\Rightarrow dq = \lambda dl$

Vedantu
Learn LIVE Online

Electric field due to a Uniformly Charged Ring



$$E_P = \int dE \cos\alpha$$

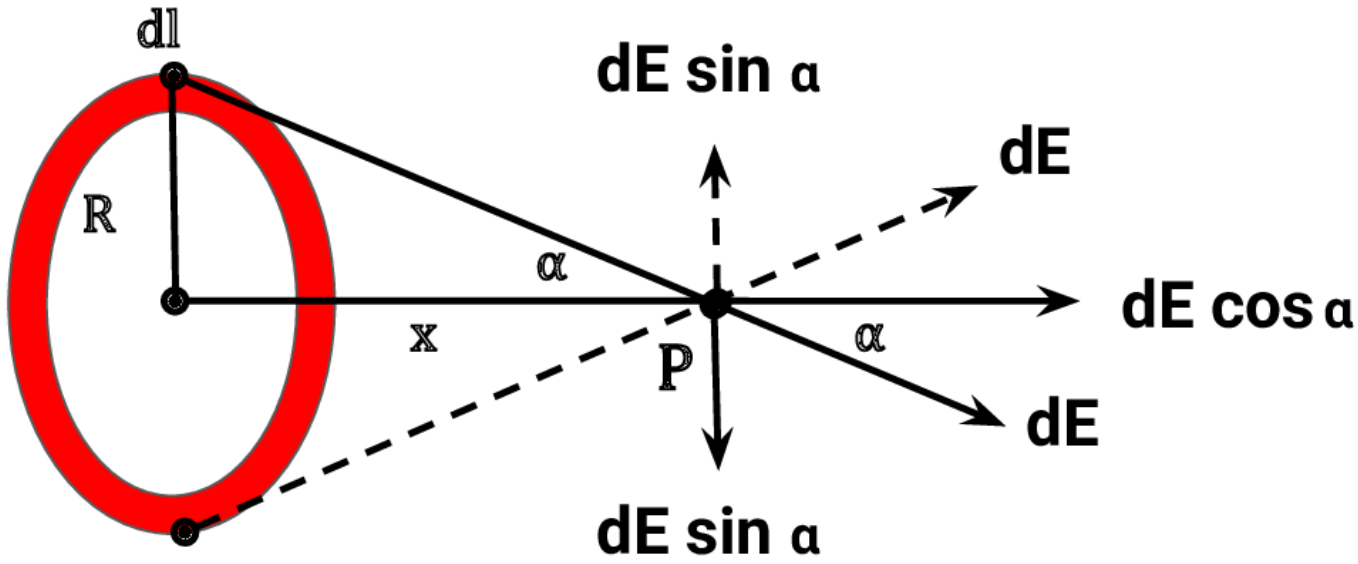
$$\Rightarrow E_P = \int_0^{2\pi R} \frac{k\lambda dl}{(R^2 + x^2)} \cos\alpha$$

Components of electric field along the y-axis will cancel out because they are equal and in opposite directions.
Components of electric field along the x-axis will add up for net electric field.

Vedantu
Learn LIVE Online

Electric field due to a Uniformly Charged Ring

Total charge on ring is +Q



From figure

$$\cos\alpha = \frac{x}{(x^2 + R^2)^{\frac{1}{2}}}$$

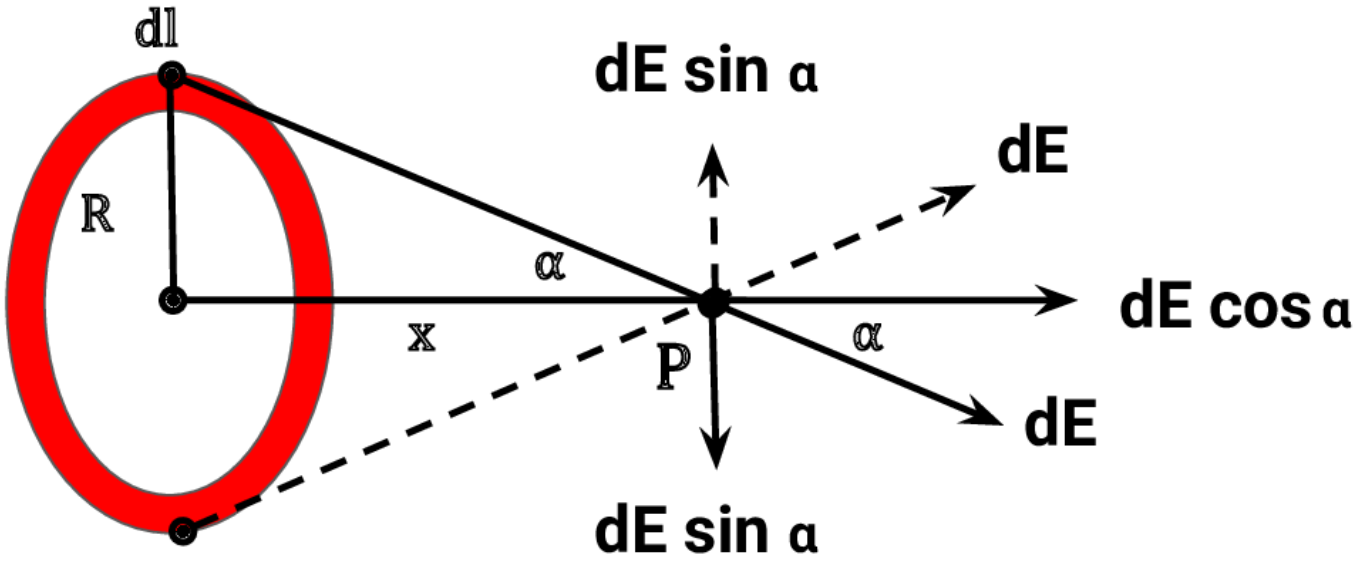
$$\Rightarrow E_P = \int_0^{2\pi R} \frac{k\lambda\, x\, dl}{(R^2 + x^2)^{\frac{3}{2}}} \quad \Rightarrow E_P = \frac{k\lambda\, x}{(R^2 + x^2)^{\frac{3}{2}}} \int_0^{2\pi R} dl$$



Vedantu
Learn LIVE Online

Electric field due to a Uniformly Charged Ring

Total charge on ring is +Q



$$\Rightarrow E_P = \frac{kx}{(R^2 + x^2)^{\frac{3}{2}}} \frac{Q}{2\pi R} [l]_0^{2\pi R}$$

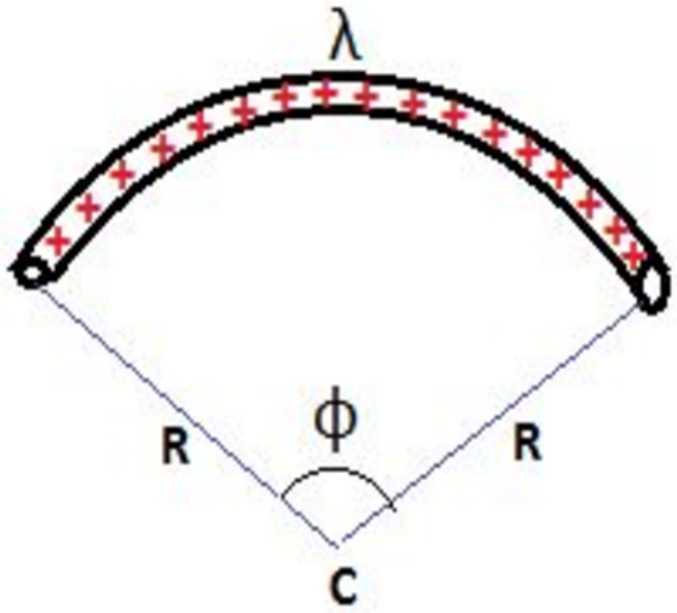
(∵ λ = Q/2πR)

$$\Rightarrow \boxed{E_P = \frac{kQ\,x}{(R^2 + x^2)^{\frac{3}{2}}}}$$

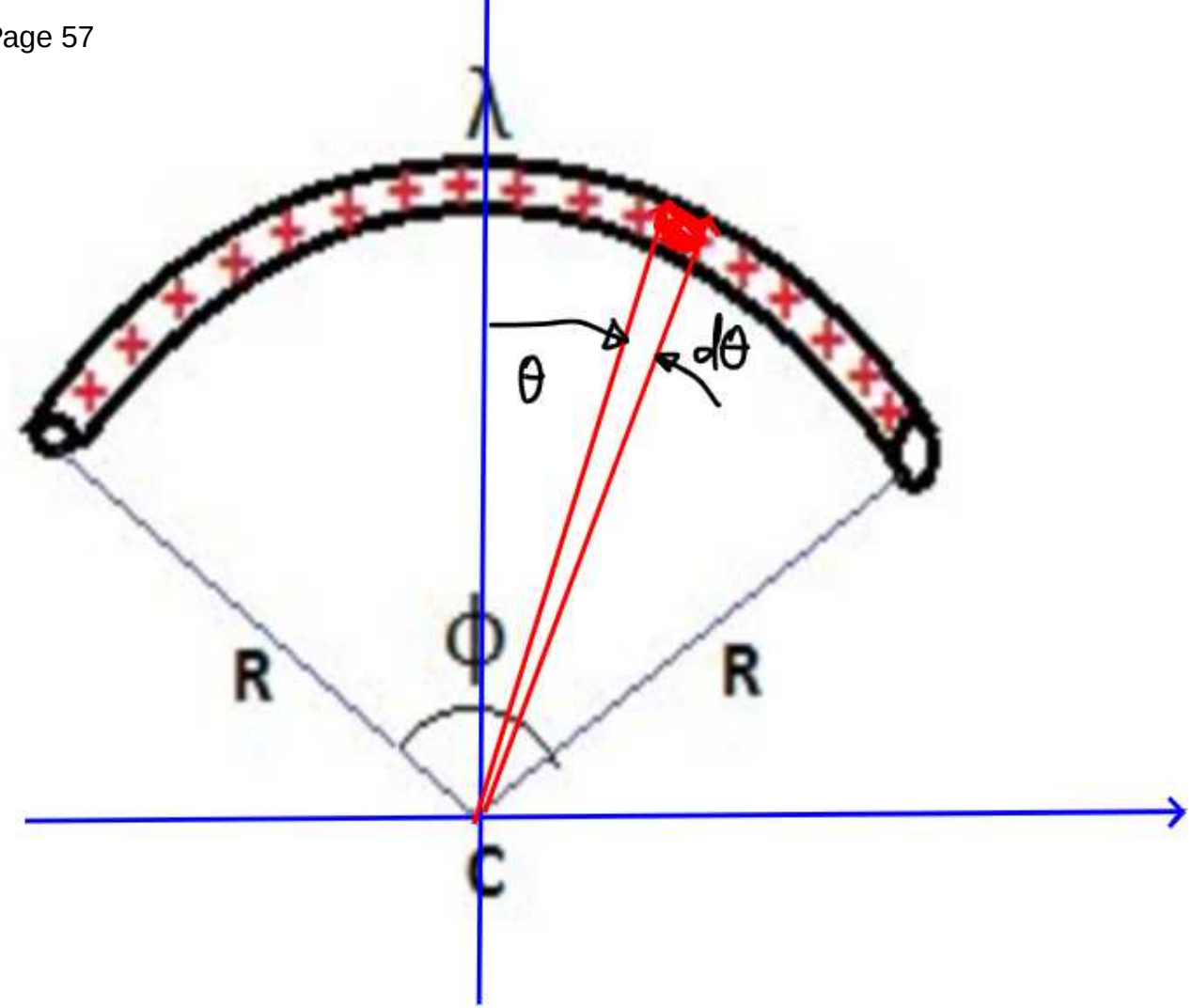
Also, $E_P = \frac{kQ}{(R^2 + x^2)} \cos\alpha$

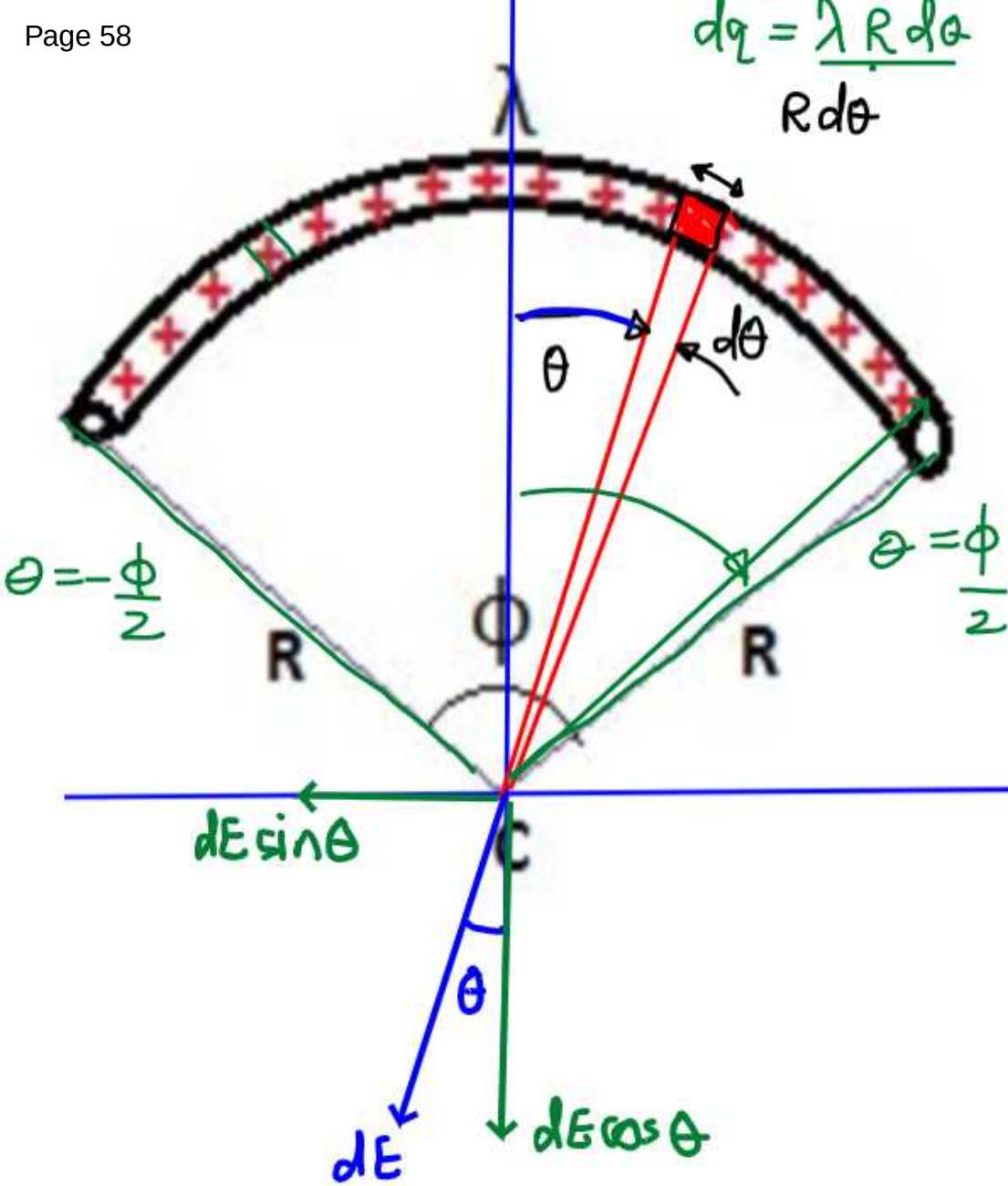
Vedantu
Learn LIVE Online

Electric field intensity at the centre of a uniformly charged arc



Vedantu
Learn LIVE Online


λ
θ
dθ
R
R
C



$$dq = \frac{\lambda R\, d\theta}{R\, d\theta}$$

$$E = \int dE \cos\theta$$

$$E = \int \frac{k\, dq}{R^2} \cos\theta$$

$$E = \frac{k}{R^2} \int \lambda R\, d\theta \cos\theta$$

$$E = \frac{k\lambda}{R} \int_{-\Phi/2}^{\Phi/2} \cos\theta\, d\theta$$

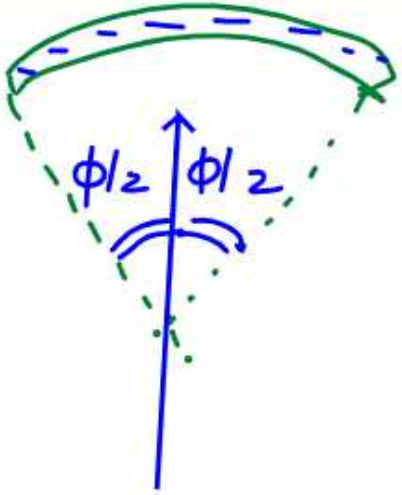
$$E = \frac{k\lambda}{R} \left[\sin\theta \right]_{-\Phi/2}^{\Phi/2}$$

$$= \frac{k\lambda}{R} \left[\sin\frac{\Phi}{2} - \sin\left(-\frac{\Phi}{2}\right) \right]$$

$$= \frac{k\lambda}{R} \left[\sin\frac{\Phi}{2} + \sin\left(\frac{\Phi}{2}\right) \right]$$

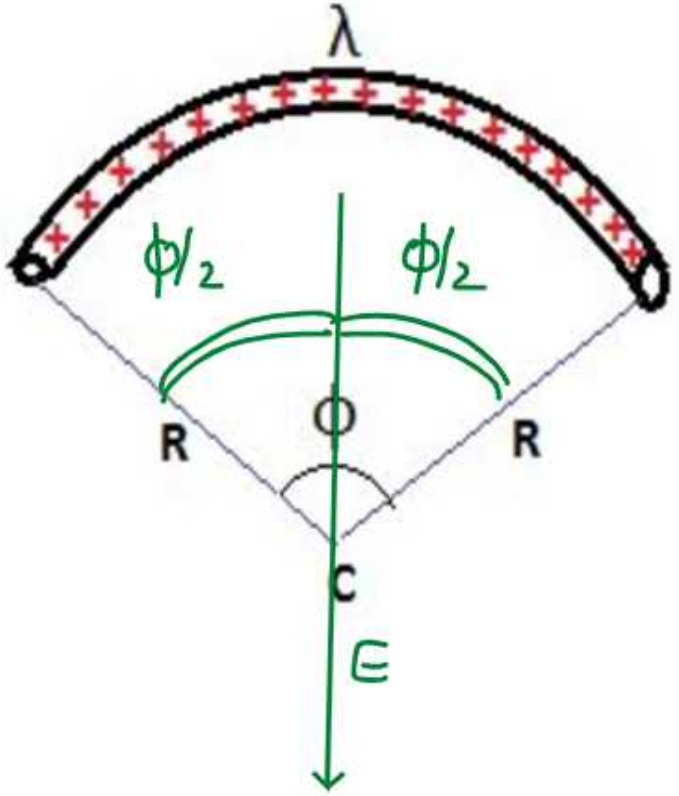
$$\boxed{E = \frac{2k\lambda}{R} \sin\left(\frac{\Phi}{2}\right)}$$

Electric field intensity at the centre of a uniformly charged arc



$$E = \frac{2k\lambda \sin\left[\frac{\phi}{2}\right]}{R}$$

$$E = \frac{2k\lambda}{R} \cdot \sin\left(\frac{\phi}{2}\right)$$



Electric field due to a Uniformly Charged arc

Chat Off

Interactivity Pod

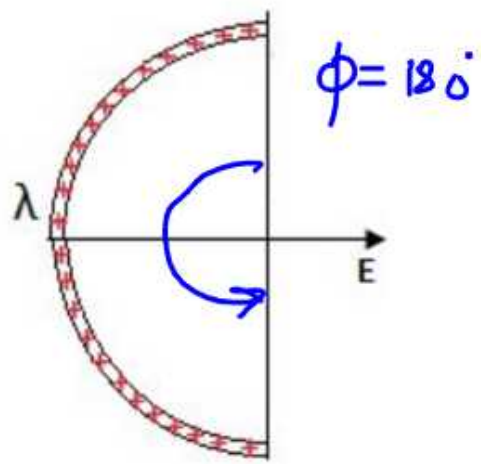
What will be the electric field at the centre of a uniformly charged semicircular arc of radius R

(A) zero

(B) $\frac{\sqrt{2}k\lambda}{R}$

(C) $\frac{2k\lambda}{R}$ ✓

(D) $\frac{k\lambda}{R}$



$$E = \frac{2k\lambda}{R} \cdot \sin\left(\frac{\phi}{2}\right)$$

$$= \frac{2k\lambda}{R} \sin\left(\frac{180^\circ}{2}\right)$$

$$= \frac{2k\lambda}{R}(1)$$

$$= \frac{2k\lambda}{R}$$

Vedantu
Learn LIVE Online

Electric field due to a Uniformly Charged arc

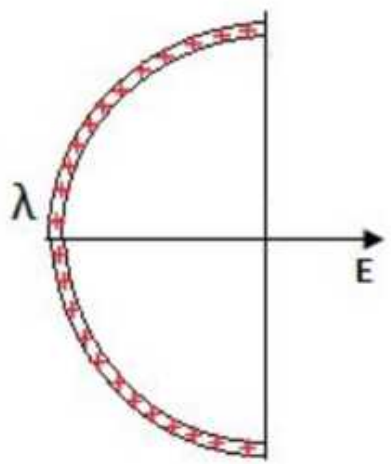
Chat On

Interactivity Pod

What will be the electric field at the centre of a uniformly charged semicircular arc of radius R

A $\frac{2k\lambda}{R}$

B zero



$$E = \frac{2k\lambda \sin\left[\frac{\phi}{2}\right]}{R}$$

$\phi = \pi/2$

So,

$$E = \frac{2k\lambda}{R}$$

Vedantu
Learn LIVE Online

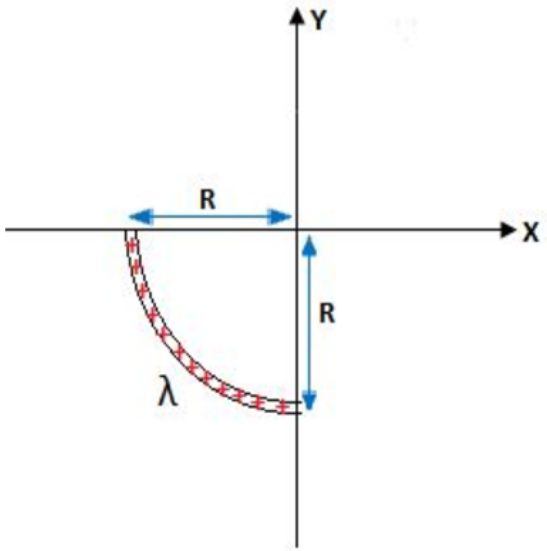
Let's Solve.



Vedantu
Learn LIVE Online

Lets Solve

Q. Find electric field intensity vector at the origin.



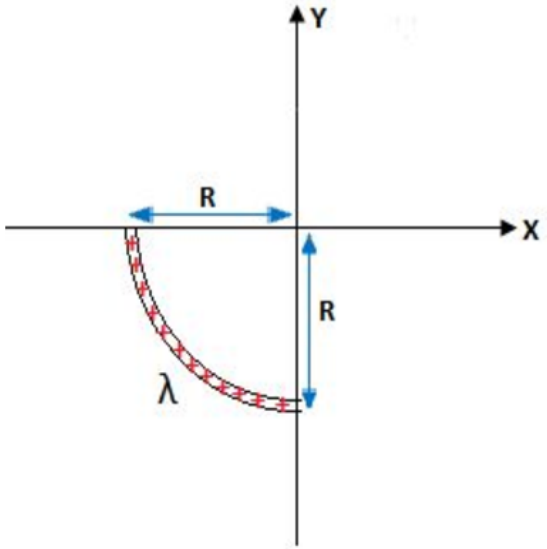
Vedantu
Learn LIVE Online

Lets Solve

Q. Find electric field intensity vector at the origin.

Soln:

$$\left[\frac{k\lambda}{R}\hat{i} + \frac{k\lambda}{R}\hat{j}\right]$$



Vedantu
Learn LIVE Online

Lets Solve

Q. Find electric field intensity vector at the origin.

Soln:

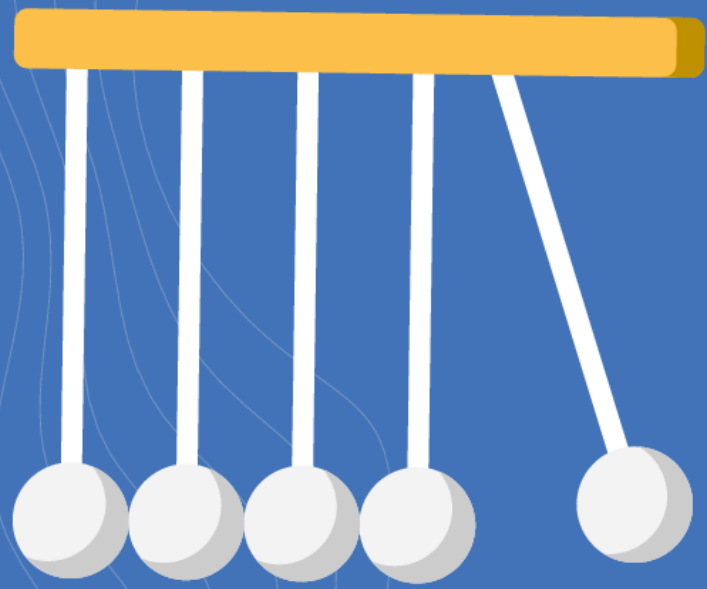
The 'E' vector at origin will be equally inclined to +ve x axis +ve y axis. Magnitude of E can be found as

$$E = \frac{2k\lambda \sin\left[\frac{\phi}{2}\right]}{R} = \frac{2k\lambda \sin\left[\frac{90^\circ}{2}\right]}{R} = \frac{\sqrt{2}k\lambda}{R}$$

Thus $\vec{E} = \frac{\sqrt{2}k\lambda}{R}[\cos 45^\circ \hat{i} + \sin 45^\circ \hat{j}] = \left[\frac{k\lambda}{R}\hat{i} + \frac{k\lambda}{R}\hat{j}\right]$

Vedantu
Learn LIVE Online

Quiz.



Vedantu
Learn LIVE Online

Quiz

Chat Off

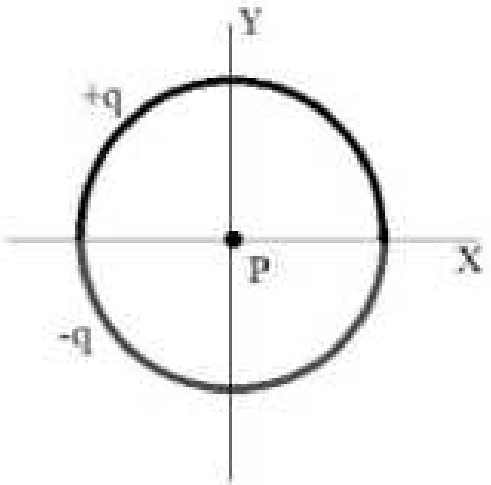
Ex. In figure a, two curved plastic rods, one of charge + q and the other of charge -q, form a circle of radius R in xy plane. The x-axis passes through their connecting points, and the charge is distributed uniformly on both rods. What is the magnitude of the electric field produced at P, the center of the circle?

A $\frac{2q}{\pi^2 R^2 \varepsilon_0}$

B $\frac{q}{\pi^2 R^2 \varepsilon_0}$

C $\frac{\sqrt{2}q}{\pi^2 R^2 \varepsilon_0}$

D Zero



Vedantu
Learn Live Online

Quiz

Chat On

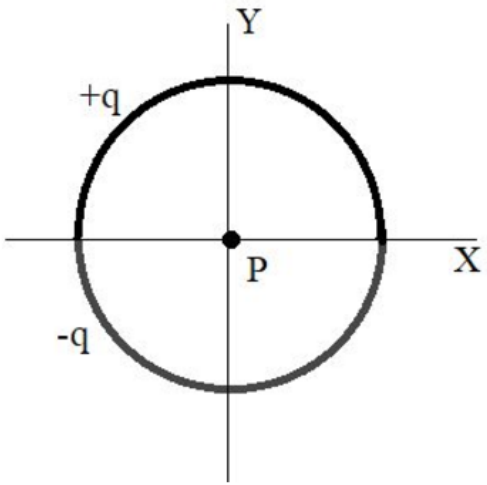
Ex. In figure a, two curved plastic rods, one of charge + q and the other of charge -q, form a circle of radius R in xy plane. The x-axis passes through their connecting points, and the charge is distributed uniformly on both rods. What is the magnitude of the electric field produced at P, the center of the circle?

A $\frac{2q}{\pi^2 R^2 \varepsilon_0}$

B $\frac{q}{\pi^2 R^2 \varepsilon_0}$

C $\frac{\sqrt{2}q}{\pi^2 R^2 \varepsilon_0}$

D Zero



Vedantu
Learn LIVE Online

Electric field intensity due to a uniformly charged straight wire/rod

Two cases:

1. Points that lie on the axis of the rod/wire. i.e. point P



2. All the other points. e.g. point T



Vedantu
Learn LIVE Online

Electric field intensity due to a uniformly charged straight wire/rod

Two cases:

1. Points that lie on the axis of the rod/wire. i.e. point P



Vedantu
Learn LIVE Online

Electric field intensity due to a uniformly charged straight wire/rod

Two cases:

2. All the other points. e.g. point T



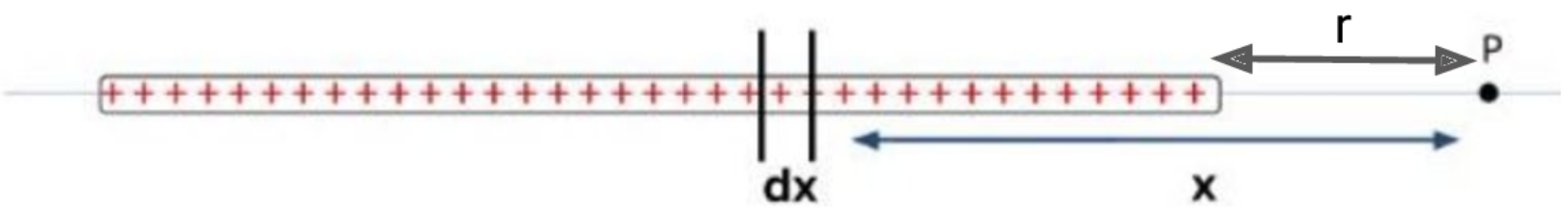
Vedantu
Learn LIVE Online

Electric field intensity due to a uniformly charged straight wire/rod

Case: 1

1. Points that lie on the axis of the rod/wire of length L, i.e. point P

Total charge on rod is Q

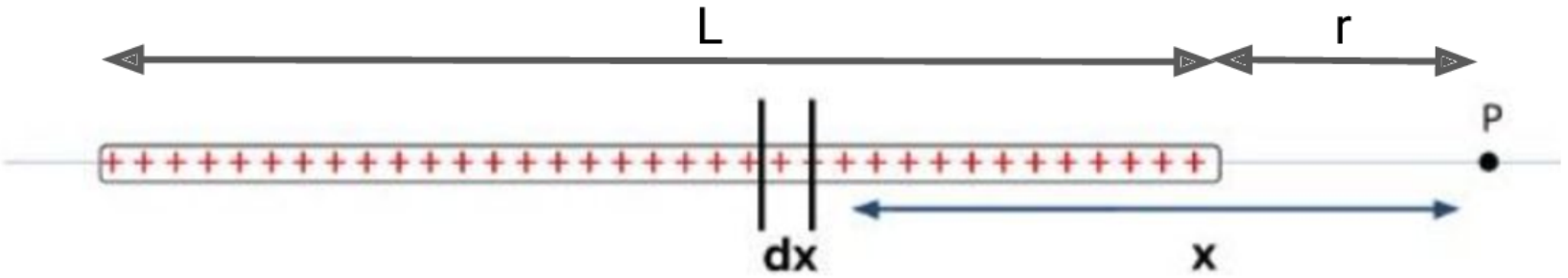


$$E_p = \frac{kQ}{r\,(r+L)}$$

Vedantu
Learn LIVE Online

Electric field intensity due to a uniformly charged straight wire/rod

Derivation:



As total charge on rod is Q,

$$dQ = \frac{Q}{L}dx$$

$$dE_p = \frac{kdQ}{x^2} = \frac{kQdx}{Lx^2}$$

$$\therefore \quad E_p = \int_r^{r+L} dE_p = \int_r^{r+L} \frac{kQdx}{Lx^2} = \frac{kQ}{L}\left[\frac{-1}{x}\right]_r^{r+L} = \frac{kQ}{L}\left[\frac{1}{r} - \frac{1}{r+L}\right] = \frac{kQ}{r(r+L)}$$

Vedantu
Learn LIVE Online

Electric field due to a Uniformly Charged arc

Chat Off

Interactivity Pod

A uniformly charged rod having 2 Coulomb charge, produces an electric field of 10^9 N/C at a point P situated at a distance of 3 m from one end of the rod. Find length of the rod.

(A) 6m

(B) 5m

(C) 2m

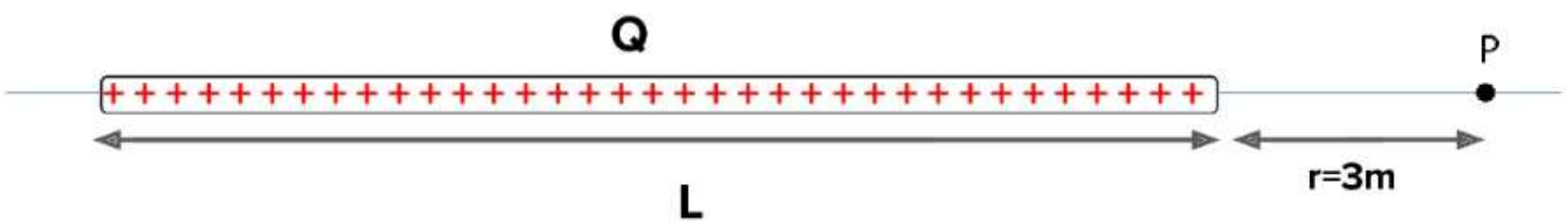
(D) 3m ✓

$$E = \frac{kQ}{r(L+r)}$$

$$10^9 = \frac{9\times10^9\times2}{3\times(L+3)}$$

$$L+3 = 6$$

$$L = 3$$



Vedantu
Learn LIVE Online

$$E = \frac{kQ}{r^2} = \frac{kQ}{r\,r} = \frac{kQ}{r(L+r)} = \frac{kQ}{r\,r} = \frac{kQ}{r^2}$$

$r >> L \;\therefore\; r + L \approx r$

Electric field due to a Uniformly Charged arc

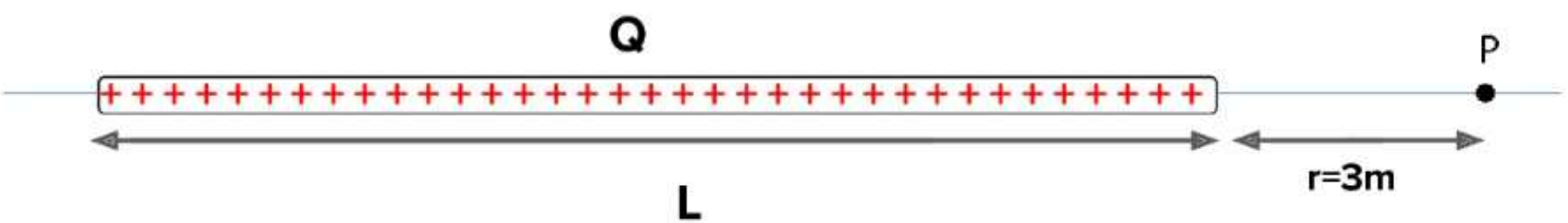
Chat On

Interactivity Pod

A uniformly charged rod having 2 Coulomb charge, produces an electric field of 10^9 N/m at a point P situated at a distance of 3 m from one end of the rod. Find length of the rod.

A 2 m

B 3 m



Vedantu
Learn LIVE Online

Electric field due to a Uniformly Charged arc

Chat On

Interactivity Pod

A uniformly charged rod having 2 Coulomb charge, produces an electric field of 10^9 N/m at a point P situated at a distance of 3 m from one end of the rod. Find length of the rod.

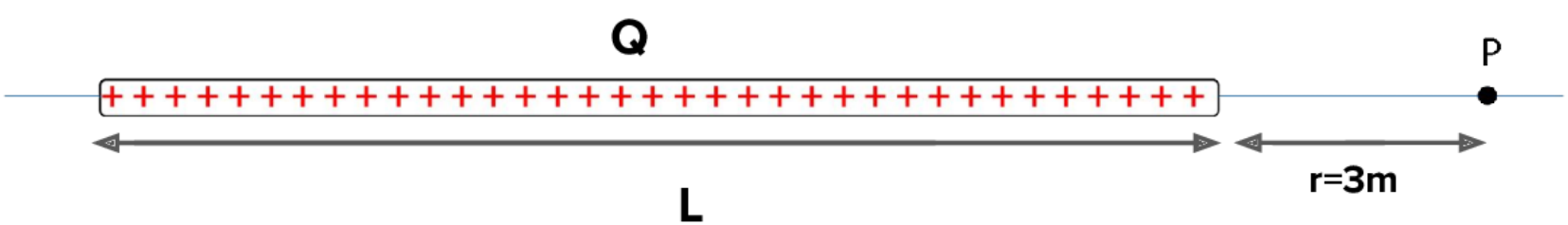
Soln:

$$E = \frac{Kq}{r(r+L)}$$

$$10^9 = \frac{9\times10^9\times2}{3(3+L)} = \frac{6\times10^9}{3+L}$$

$$\Rightarrow 3+L=6$$

$$\Rightarrow L = 3\,m$$



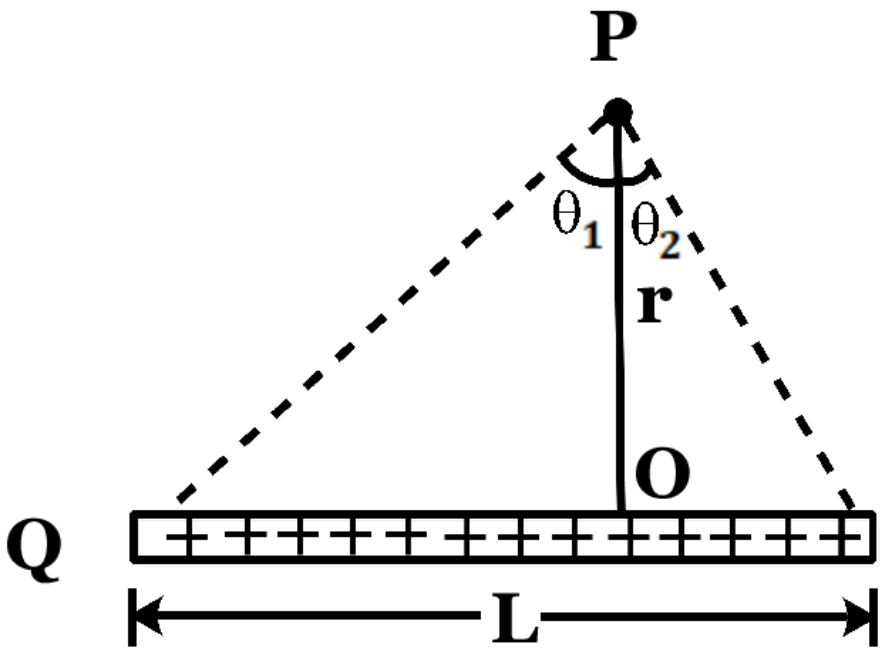
Vedantu
Learn LIVE Online

Electric Field due to Uniformly charged Rod

So electric field at a point P situated at a distance r from a uniformly charged rod and making angles θ_1 and θ_2 is

$$\Rightarrow E_{||} = E_x = \frac{k\lambda}{r}(\cos\theta_2 - \cos\theta_1)$$

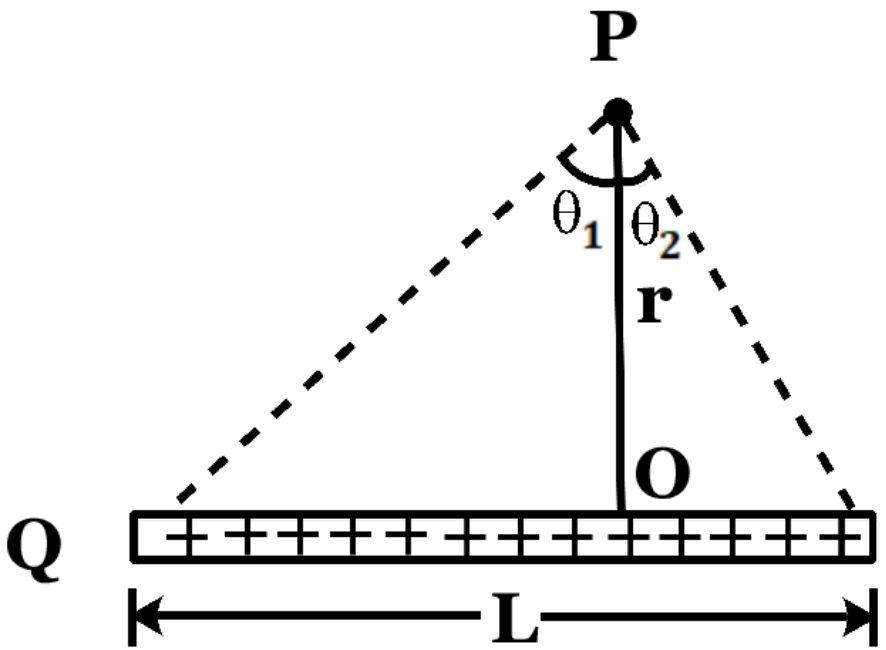
$$\Rightarrow E_{\perp} = E_y = \frac{k\lambda}{r}(\sin\theta_1 + \sin\theta_2)$$



Vedantu
Learn LIVE Online

Electric Field due to Uniformly charged Rod

So electric field at a point P situated at a distance r from a uniformly charged rod and making angles θ_1 and θ_2 is



Vedantu
Learn LIVE Online

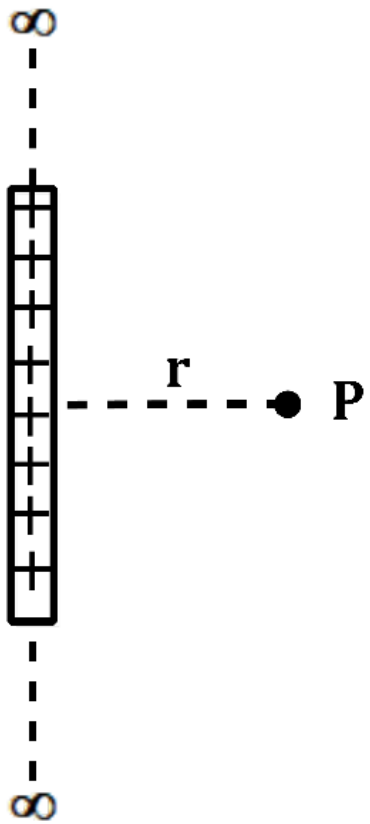
Electric Field due to Uniformly charged Rod

For infinite length of rod (wire),

$$\theta_1 = \theta_2 = \frac{\pi}{2}$$

$$\Rightarrow \vec{E}_{||} = \frac{k\lambda}{r}(\cos 90° - \cos 90°)$$

$$\Rightarrow \vec{E}_{||} = 0$$



Vedantu
Learn LIVE Online

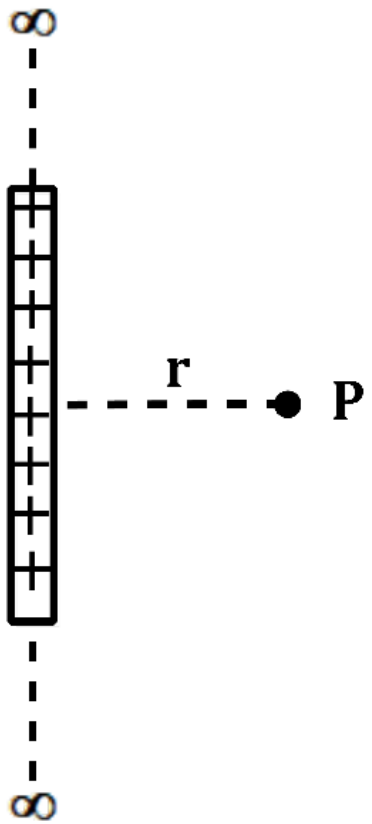
Electric Field due to Uniformly charged Rod

For infinite length of rod (wire),

$$\vec{E}_{\perp} = \frac{k\lambda}{r}(\sin\theta_1 + \sin\theta_2)$$

$$\Rightarrow \vec{E}_{\perp} = \frac{k\lambda}{r}(\sin 90° + \sin 90°)$$

$$\Rightarrow \vec{E}_{\perp} = \frac{2k\lambda}{r}$$



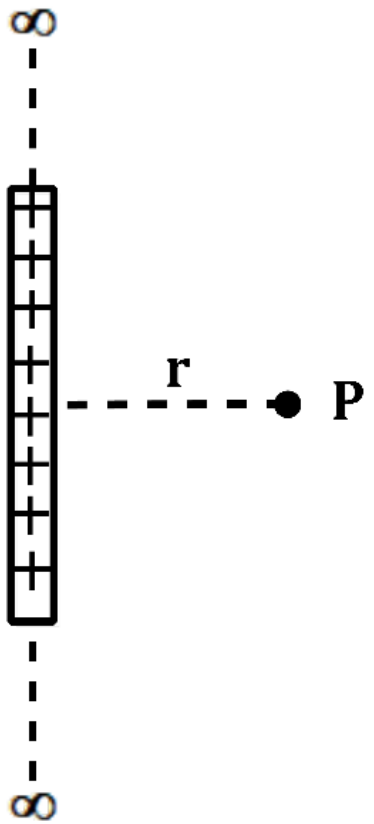
Vedantu
Learn LIVE Online

Electric Field due to Uniformly charged Rod

so, $\vec{E}_{net} = \vec{E}_{\parallel} + \vec{E}_{\perp}$

$$= 0 + \frac{2k\lambda}{r}$$

$$= \frac{2k\lambda}{r}$$ (In perpendicular direction)



Vedantu
Learn LIVE Online

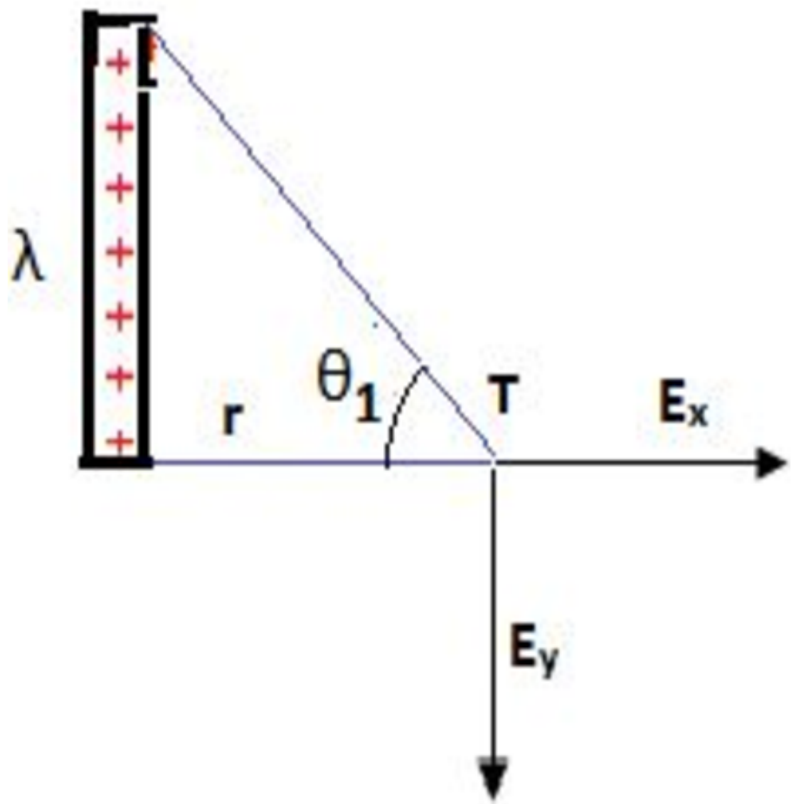
Electric Field due to Uniformly charged Rod

Special cases :

Uniformly charged semi-infinite wire/rod

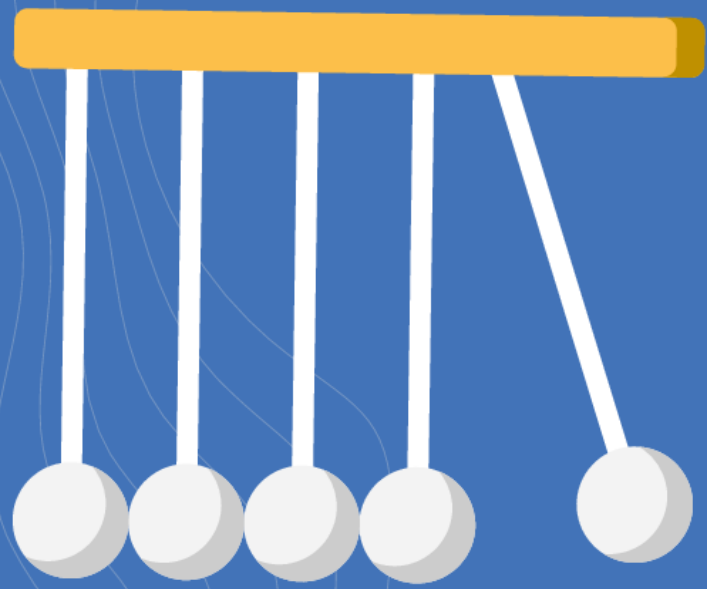
$$|E_{\parallel}| = \frac{k\lambda}{r}$$

$$|E_{\perp}| = \frac{k\lambda}{r}$$



Vedantu
Learn LIVE Online

Quiz.



Vedantu
Learn LIVE Online

Quiz

Chat Off

Ex. Find the electric field due to a semi infinite wire, at a perpendicular distance r from it's one end, having linear charge density λ.

A $\frac{k\lambda}{r}$

B $\frac{\sqrt{2}k\lambda}{r}$

C $\frac{2k\lambda}{r}$

D $\frac{1}{2\sqrt{2}}\frac{k\lambda}{r}$

Vedantu
Learn Live Online

Quiz

Chat On

Ex. Find the electric field due to a semi infinite wire, at a perpendicular distance r from it's one end, having linear charge density λ.

A $\frac{k\lambda}{r}$

B $\frac{\sqrt{2}k\lambda}{r}$

C $\frac{2k\lambda}{r}$

D $\frac{1}{2\sqrt{2}}\frac{k\lambda}{r}$

Vedantu
Learn LIVE Online

Quiz

Chat On

Ex. Find the electric field due to a semi infinite wire, at a perpendicular distance r from it's one end, having linear charge density λ.

Soln:

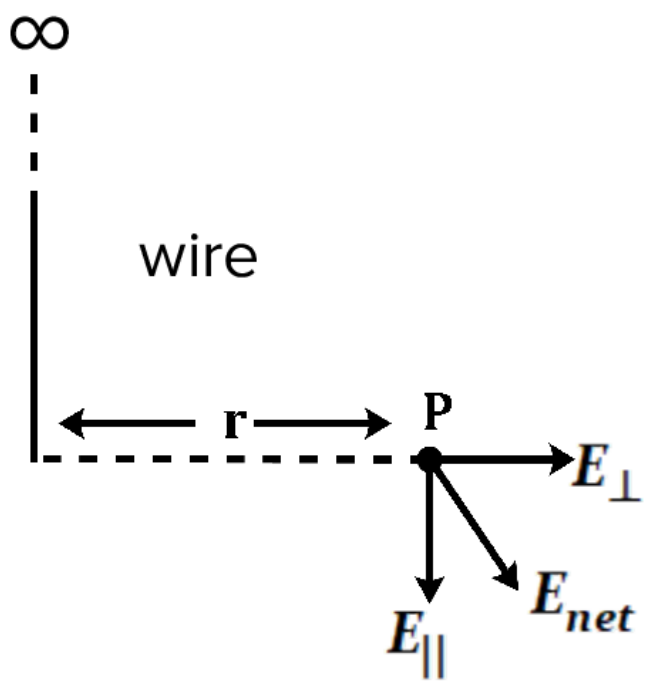
For the given situation,

$$\theta_1 = 90°, \theta_2 = 0$$

So, $$E_\perp = \frac{k\lambda}{r}(\cos 0° - \cos 90°)$$

$$\Rightarrow E_\perp = \frac{k\lambda}{r}$$

Now $$\Rightarrow E_{||} = \frac{k\lambda}{r}(\sin 0° + \sin 90°)$$



Vedantu
Learn LIVE Online

Quiz

Chat On

Ex. Find the electric field due to a semi infinite wire, at a perpendicular distance r from it's one end, having linear charge density λ.

Soln:

$$= \frac{k\lambda}{r}$$

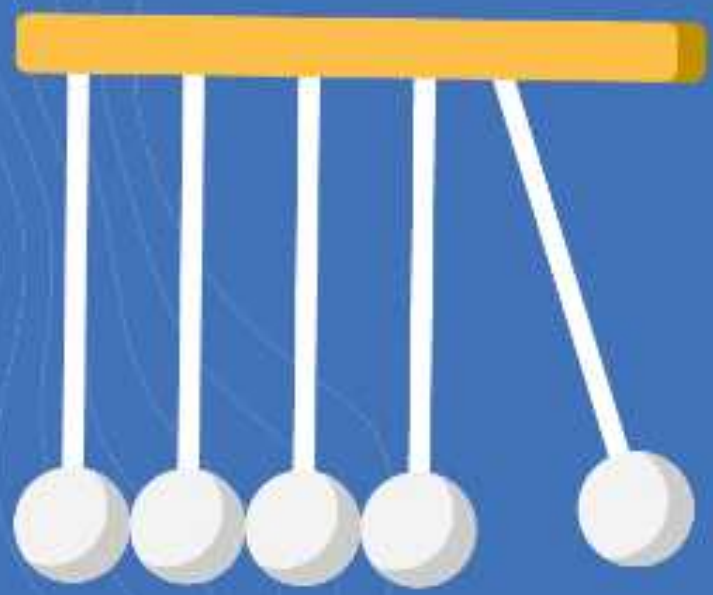
So, net electric field at point P

$$E_{net} = \sqrt{(E_{||})^2 + (E_{\perp})^2 + 2(E_{\perp})(E_{||}) \cos 90°}$$

$$E_{net} = \frac{\sqrt{2}k\lambda}{r}$$

Vedantu
Learn LIVE Online

Quiz.



Quiz

Chat Off

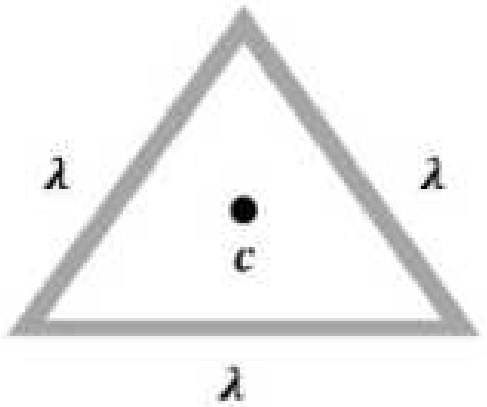
Ex. Given an equilateral triangle with side λ .Find electric field strength at the centroid. The linear charge density is as shown in figure.

A $\frac{3\lambda}{\pi\varepsilon_0 l}$

B Zero

C $\frac{\sqrt{2}\lambda}{\pi\varepsilon_0 l}$

D $\frac{\lambda}{\pi\varepsilon_0 l}$



Vedantu
Learn Live Online

Quiz

Chat On

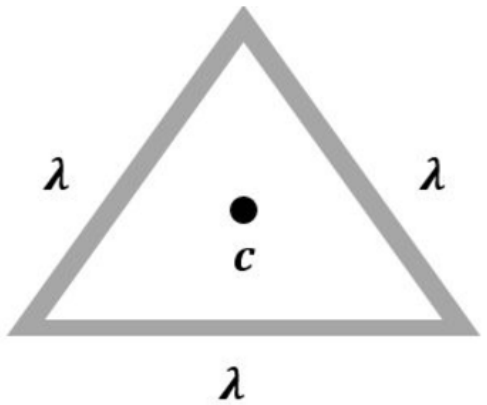
Ex. Given an equilateral triangle with side λ .Find electric field strength at the centroid. The linear charge density is as shown in figure.

A $\frac{3\lambda}{\pi\varepsilon_0 l}$

B Zero

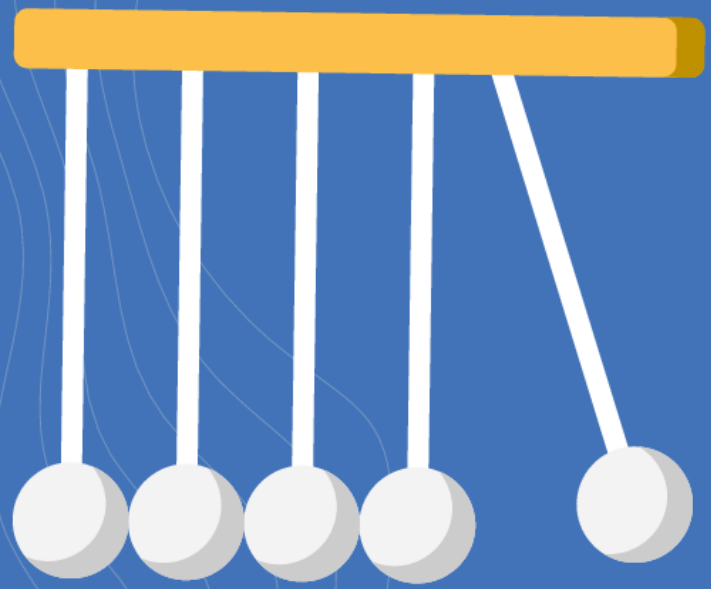
C $\frac{\sqrt{2}\lambda}{\pi\varepsilon_0 l}$

D $\frac{\lambda}{\pi\varepsilon_0 l}$



Vedantu
Learn LIVE Online

Quiz.



Vedantu
Learn LIVE Online

Quiz

Chat Off

Ex. A semi infinite line charge having a constant linear charge density 5nC/m lies along the x-axis from x =-∞ to 0. Find the electric field at point P (0, 5).

A. $9\sqrt{2}$ N/C making an angle 90° with X-axis

B. 9 N/C making an angle 90° with X-axis

C. $9\sqrt{2}$ N/C making an angle 45° with X-axis

D. 9 N/C making an angle 45° with X-axis

Vedantu
Learn Live Online

Quiz

Chat On

Ex. A semi infinite line charge having a constant linear charge density 5nC/m lies along the x-axis from x =-∞ to 0. Find the electric field at point P (0, 5).

A. $9\sqrt{2}$ N/C making an angle 90° with X-axis

B. 9 N/C making an angle 90° with X-axis

C. $9\sqrt{2}$ N/C making an angle 45° with X-axis

D. 9 N/C making an angle 45° with X-axis

Chat On

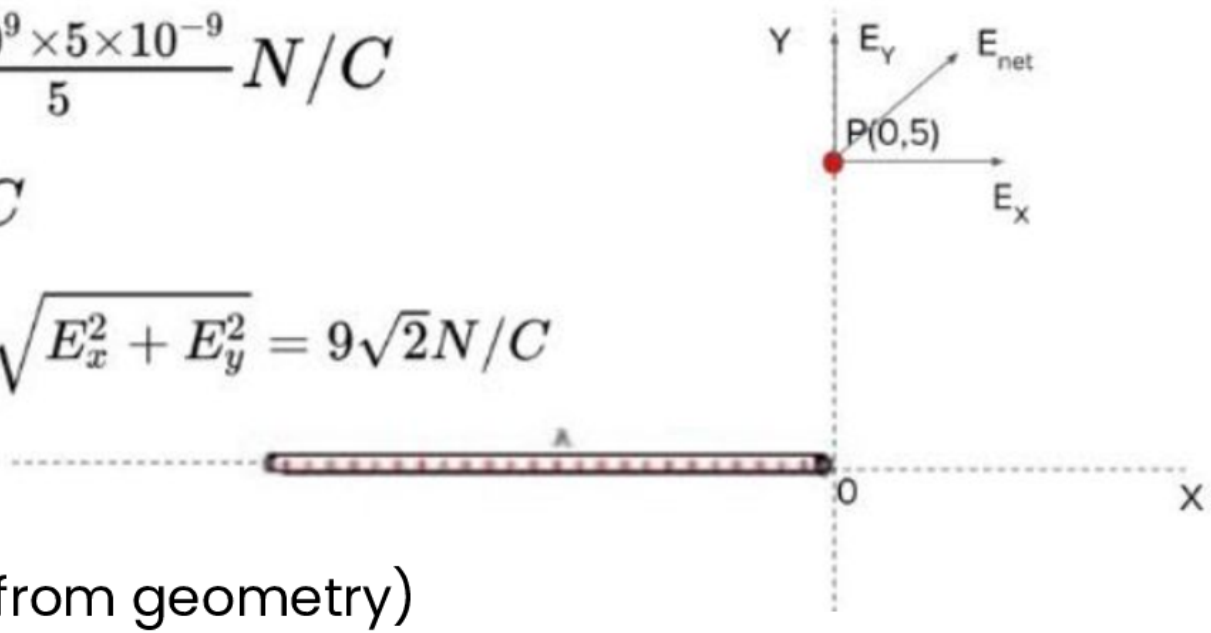
Ex. A semi infinite line charge having a constant linear charge density 5nC/m lies along the x-axis from x =-∞ to 0. Find the electric field at point P (0, 5).

Soln:

$$E_x = \frac{k\lambda}{r} = \frac{9\times10^9\times5\times10^{-9}}{5} N/C$$

$$E_y = \frac{k\lambda}{r} = 9N/C$$

$$E_x \perp E_y \Rightarrow E_{net} = \sqrt{E_x^2 + E_y^2} = 9\sqrt{2}N/C$$

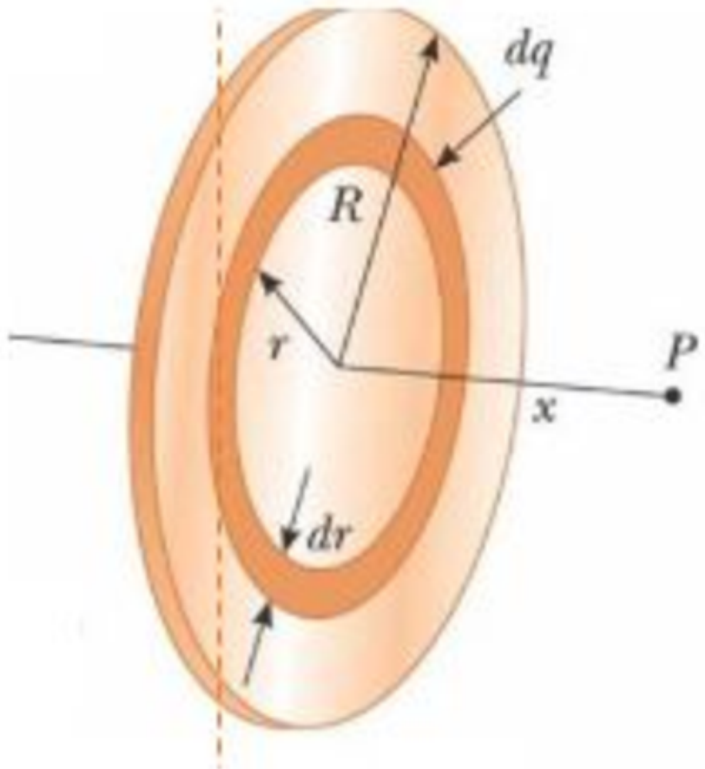


(Direction is clear from geometry)

Vedantu
Learn LIVE Online

Electric field intensity due to a uniformly charged Disc

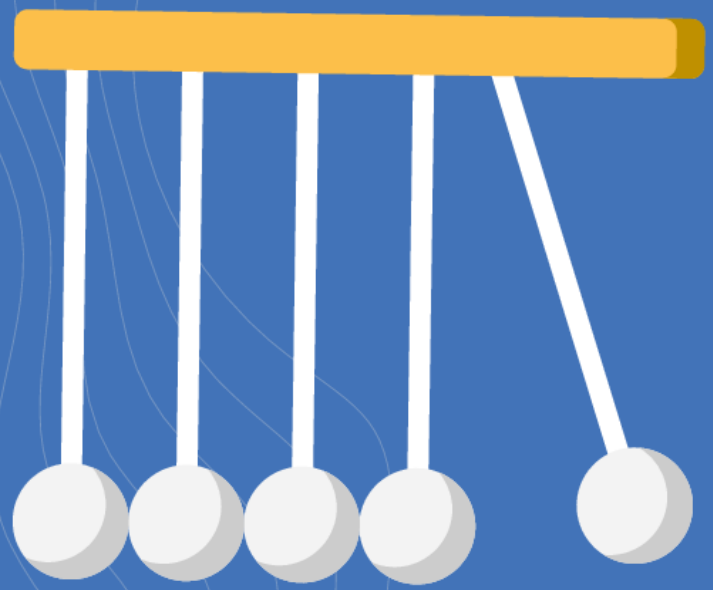
$$E = \frac{\sigma}{2\varepsilon_o}\left[1 - \frac{x}{\sqrt{x^2+R^2}}\right]$$





Vedantu
Learn LIVE Online

Quiz.



Vedantu
Learn LIVE Online

Quiz

Chat Off

Ex. A charged disc of radius 3 m has charge density σ. Electric field at a distance of 4 m from its centre on axis is

A $\frac{3\sigma}{5\epsilon_0}$

B $\frac{\sigma}{10\epsilon_0}$

C $\frac{\sigma}{8\epsilon_0}$

D None

Vedantu
Learn Live Online

Quiz

Chat On

Ex. A charged disc of radius 3 m has charge density σ. Electric field at a distance of 4 m from its centre on axis is

A $\frac{3\sigma}{5e_0}$

B $\frac{\sigma}{10e_0}$

C $\frac{\sigma}{8e_0}$

D None

Vedantu
Learn LIVE Online

Ex. A charged disc of radius 3 m has charge density σ. Electric field at a distance of 4 m from its centre on axis is

Soln:

$$E_x = \frac{\sigma}{2\varepsilon_0}\left[1 - \frac{x}{\sqrt{x^2+R^2}}\right]; \quad \begin{matrix} x = 4m \\ R = 3m \end{matrix}$$

$$E_x = \frac{\sigma}{2\varepsilon_0}\left[1 - \frac{4}{\sqrt{16+9}}\right] = \frac{\sigma}{2\varepsilon_0}\left(1 - \frac{4}{5}\right)$$

$$E_x = \frac{\sigma}{2\varepsilon_0}\frac{1}{5}$$

$$E_x = \frac{\sigma}{10\varepsilon_0}$$

Electric field

① Due to a point charge



$$E \text{ at } p = \frac{kQ}{r^2}$$

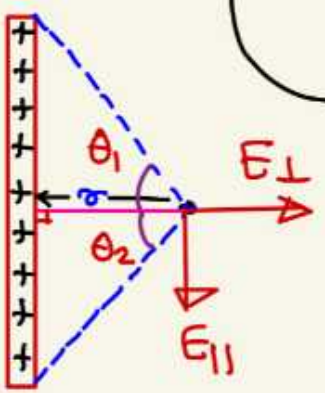
② Due to rod

→ (I) Point on axis



$$E \text{ at } p = \frac{kQ}{r(L+r)}$$

→ (II) Point Not on axis



$$E_{11} = \frac{k\lambda}{r}[\cos\theta_2 - \cos\theta_1]$$

$$E_{\perp} = \frac{k\lambda}{r}[\sin\theta_2 + \sin\theta_1]$$

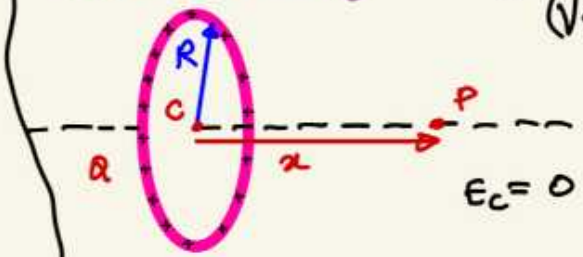
③ Due to arc



$$E = \frac{2k\lambda}{R}\sin\left(\frac{\phi}{2}\right)$$

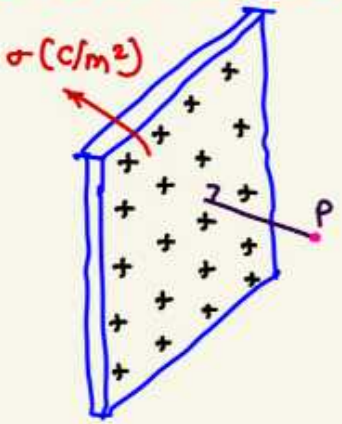
④ Due to Ring

$$E_p = \frac{kQx}{(\sqrt{x^2+R^2})^3}$$



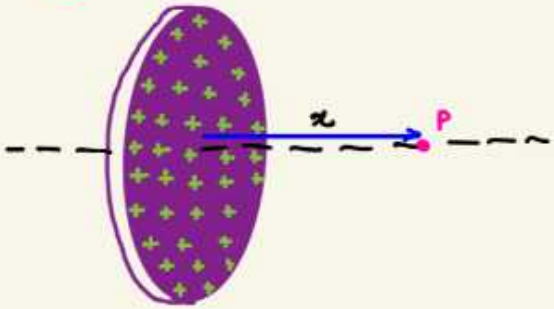
$$E_c = 0$$

⑤ Due infinite sheet



$$E_p = \frac{\sigma}{2\varepsilon_0}$$

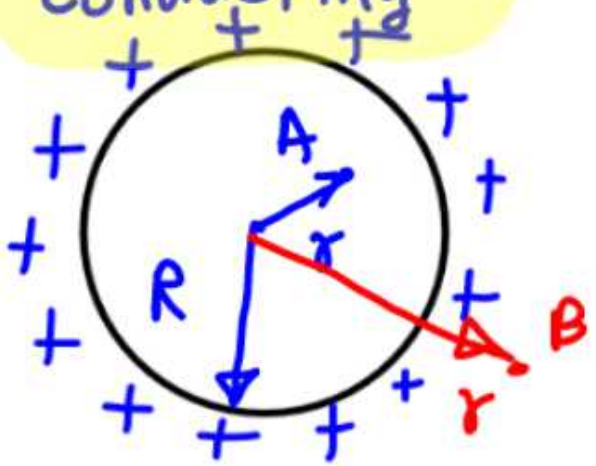
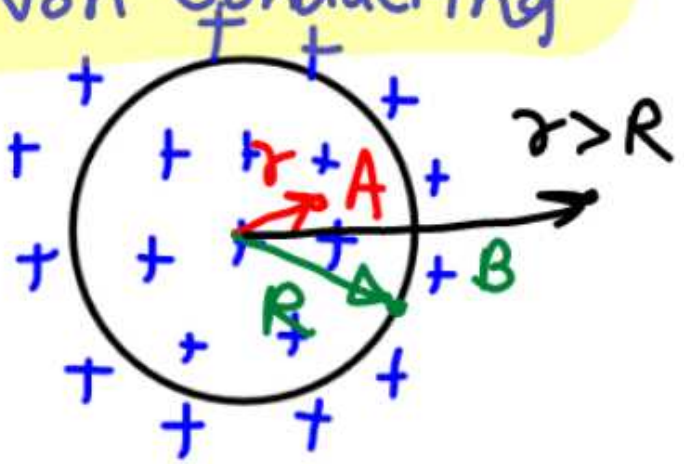
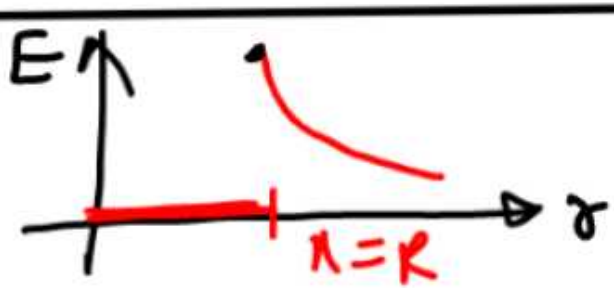
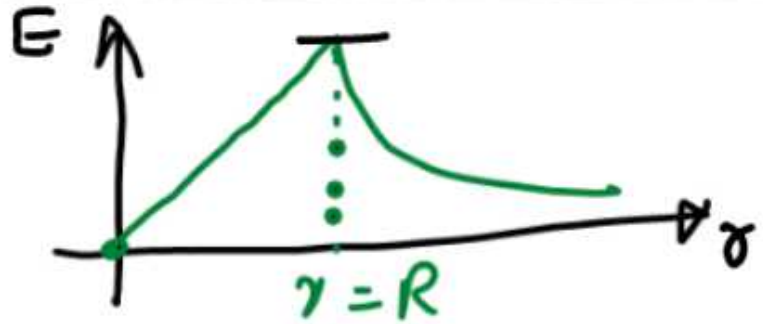
⑥ Disc



$$E_p = \frac{\sigma}{2\varepsilon_0}\left[1 - \frac{x}{\sqrt{R^2+x^2}}\right]$$

→ Infinitly long rod, $E_{11} = 0$, $E_{\perp} = \frac{2k\lambda}{r}$] → $E_{Net} = \frac{2k\lambda}{r}$

→ Semi Infinite rod $E_{11} = E_{\perp} = \frac{k\lambda}{r}$] → $E_{Net} = \frac{\sqrt{2}k\lambda}{r}$

r	Conducting	Non conducting
		
r<R inside	$E = 0$	$E = \frac{KQr}{R^2R} = \frac{KQr}{R^3}$ $E \propto r$
r=R on	$E = \frac{KQ}{R^2}$	$E = \frac{KQ}{R^2}$
r>R outside	$E = \frac{KQ}{r^2}$ $E \propto \frac{1}{r^2}$	$E = \frac{KQ}{r^2}$ $E \propto \frac{1}{r^2}$
Graph		

$k = 9 \times 10^9$

Summary.



Vedantu
Learn LIVE Online

Summary.

- Force due to multiple charges and Vector form of Coulomb's law and Role of dielectric.
- Introduction to Electric field and Electric field due to point charge
- Electric field due to a system of charges and superposition principle

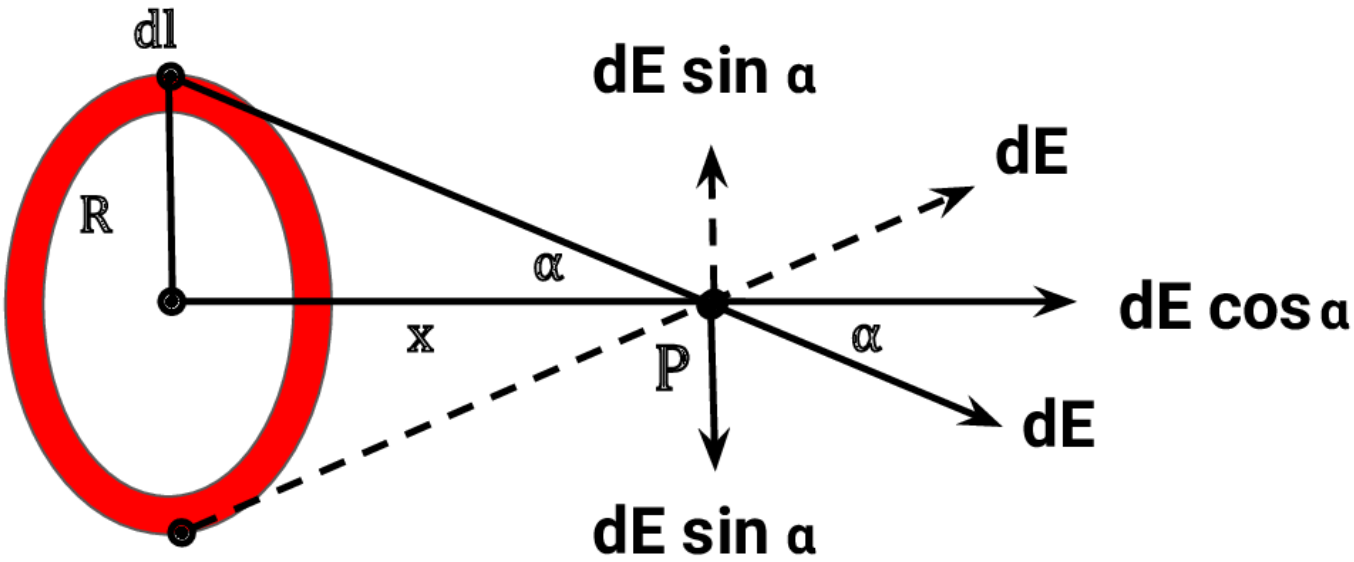
Vedantu
Learn LIVE Online

Summary.

➔ Electric field due to a ring

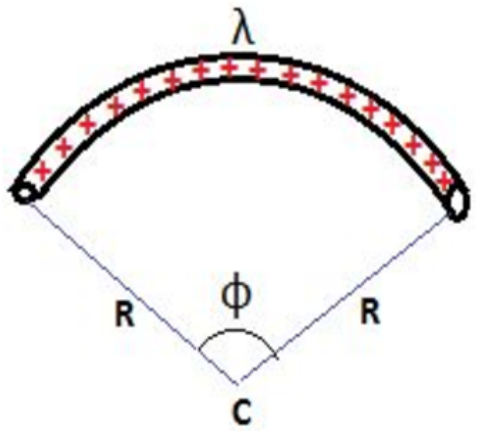
$$E_P = \frac{kQ\,x}{(R^2 + x^2)^{\frac{3}{2}}}$$

Total charge on ring is +Q



➔ Electric field due to a arc

$$E = \frac{2k\lambda \sin\left[\frac{\phi}{2}\right]}{R}$$



Vedantu
Learn LIVE Online



HW

EL-2 - 16 to 21

EL-3 - 1 to 5, 10, 11, 14 to 18, 20, 21

Thank You.

Share your feedback
it matters.